Alliance FTP Security

Page Information

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Introduction

FTP (File Transfer Protocol) is one of the most commonly used applications to transfer files on TCP/IP networks like the Internet. Every IBM i shipped by IBM includes the FTP application and it is commonly activated to provide file transfer support. It is a powerful application. In addition to providing the ability to upload or download a file, the FTP application lets you:

- Create a new library
- Delete a library
- List the contents of a library
- Delete a file or many files in a library
- Execute IBM i commands

As you can see FTP provides extraordinary capabilities to clients who access the IBM i, perhaps more capability than you want them to have.

The IBM i FTP application is based on native IBM i security mechanisms. This means that no one can access the FTP application without providing a valid IBM i user ID and password. Unfortunately, that is the extent of the security offered by IBM in the FTP application. Once a user has access to the FTP application, they can perform any of the functions above. This represents a dangerous open door to the IBM i, especially if your IBM i is connected to the Internet.

Alliance FTP Security (Alliance) was designed to help the IBM i system administrator control who gets access to the IBM i FTP application, and what they can do when they are allowed to use FTP. In addition, Alliance will keep detailed logs on all FTP activity by users and will assist in problem analysis and activity audit. You can selectively give users access to any of the FTP functions and can monitor their activities. If you enable the Intruder Alert feature of Alliance

FTP you can proactively protect your IBM i from break in attempts. Alliance FTP will send you alert messages and will even terminate FTP support if you request it. Alliance FTP is a powerful tool to help you control and monitor FTP activity on the IBM i.

IBM i FTP Security

Overview

The IBM FTP application implements user ID and password controls for access to the IBM i. The user ID entered must be for a valid user profile on the IBM i. The password must be valid for the user profile. This means that no user ID can be longer that 10 characters. The user ID and password are not secure when sent over the network. If you are working on the Internet you need to be aware of the fact that the user ID and password may not be secure and encrypted. Once a user is logged on to the IBM i all normal FTP commands are available to the user. No other security is available to control users of the FTP application.

Many IBM i customers secure their IBM is and applications by implementing a menu system for user access. A user only sees applications and databases to which they are authorized and for which they have a legitimate business need. It is important to realize that no such controls exist for an FTP user. If you have relied on a menu system to control access to the IBM i, it is likely that an FTP user will have inappropriate access to applications and data on your IBM i.

Consider the following about a user logged on to the IBM i FTP application:

- An FTP user can use the LS and DIR commands to view the contents of any library to which they have Read access. After listing a library they can retrieve any database file to which they have read access. This might well include financial database files on your system.
- An FTP user can upload files to the IBM i and replace existing files if they have rights to the files. The uploaded file can be a save file that contains programs the user has written. These programs could produce unwelcome results on your system.
- An FTP user can execute any IBM i command using the `RCMD` subcommand facility in FTP. If authorized to the commands the FTP user can power down the system, clear libraries, upload save files, replace programs, and perform many other functions that may endanger your system.

Exit points

In order to allow customers to develop additional security for FTP applications, IBM provided “exit points” in the FTP application. An exit point is simply a program call to an external application that can provide additional security controls. However, IBM did not create the programs to actually provide that additional level of control. That is what Alliance does for you. Alliance takes advantage of the FTP application exit points to provide additional security controls and audit trails not provided by IBM.

There are three exit points used by Alliance:

- FTP Logon Validation: QIBM_QTMF_SVR_LOGON
- FTP Client Request Validation: QIBM_QTMF_CLIENT_REQ

- FTP Server Request Validation: QIBM_QTMF_SERVER_REQ

The first exit point is called every time a user logs on to FTP on the IBM i. The Alliance exit point program **`ATC100`** provides security validation.

The last two exit points both handle request validation and are processed by the same Alliance program **`ATC200`** . When you activate Alliance you are registering the Alliance programs to the IBM FTP software and Alliance will be called by the exit points for every action performed by a user.

Anonymous user support

The IBM i FTP application recognizes the ANONYMOUS user ID as a special case during log in. If you access FTP and enter ANONYMOUS as the user ID you will be prompted to enter your e-mail address. However, there are no other differences in how anonymous users are handled. IBM recommends creating a user profile named ANONYMOUS with no password and then creating an exit point program to control what this user can do.

Alliance will take care of all of these tasks for you and provide additional controls over library, FTP command, and remote command access. In addition, Alliance allows you to specify any user profile you want as the profile to be used by anonymous users. An actual password is provided during anonymous log on and more appropriate controls are in place. See the section below for an additional discussion of anonymous FTP access using Alliance.

Alliance FTP Security

That which is not expressly permitted is denied

Almost all security firewall systems work on the principle that “That which is not expressly permitted is denied.” That means a system administrator must give a user access to an application or function before it can be used. Alliance helps you implement this type of policy by allowing you to specify which FTP functions a person can use when logged on with FTP. If you only want the user to be able to retrieve a file in the QGPL library, you can implement this policy and prohibit the user from uploading files, executing system commands, or any other FTP function. You can restrict access to a known list of users, to a known set of client IP network addresses, and to specific libraries on the system. And you can give users access to only those FTP functions that they need.

Controlling user access by name

You can give the ability to use FTP to specific IBM i users and exclude all other IBM i users. In this manner you can be sure that only those users who have a need for FTP functions are actually able to use FTP. The ability to limit this capability becomes more important as PC operating systems proliferate with built-in support for TCP/IP and FTP. Once a user has a profile and password, they can log on to the IBM i with FTP. Alliance gives you the ability to exclude any valid IBM i user who has not been authorized by the system administrator to FTP functions.

Controlling access for unregistered users

With Alliance you have an option to allow or exclude unregistered users from IBM i FTP. A registered user is one you have defined with the Work With FTP Users option on the Alliance main menu. An unregistered user is one that is not defined to Alliance. It is recommended that you not allow access to unregistered users as this provides an opportunity for unwelcome use of FTP on your system, but you do have the ability to allow unregistered users access to FTP if you need to. An unregistered user is allowed to pass through the Alliance FTP facility, but they must still have a user profile and valid password for the IBM i. Alliance does not disable any normal IBM i security mechanism for unregistered users.

Controlling user access by IP address

You can also control access to FTP based on the user's IP address. This includes the ability to control access based on the entire address, or just the network portion of the address. For example, a user named Joe Smith has a PC with the address of 198.241.155.164. You can authorize that this address only be allowed to use FTP on the IBM i, and other addresses are to be rejected. This provides an additional level of security in case a user profile and password are captured by unwelcome users.

In addition to the ability to specify a specific IP address, you can define just the portion of the address that represents the network address, or subnetworks. For example, you may know that all users in New York have IP addresses that start with "198.241." You can enable anyone with an address that starts like this to use FTP and exclude others. In this way you can authorize entire workgroups for FTP access.

Please note that IP address control may not work for you if you are using dynamic IP address assignment for nodes on your network. Also, users who connect to the IBM i via the Internet will probably have a different IP address each time if they use a dial connection. These are typically assigned dynamically by the Internet Service Provider (ISP) during logon.

Controlling user access to FTP functions

One you have authorized a user for access to FTP, you can restrict their functions to specific FTP operations. For example, the FTP facility defines this group of capabilities within FTP:

- Create a directory (library)
- Delete a library
- Change the default library
- List files in a library
- Delete files in a library
- Rename files in a library
- Upload a file to a library
- Download a file from a library
- Execute a remote command on the IBM i

Controlling user access to libraries

Under normal FTP security an FTP user can access any library and file to which they have read authority. In most IBM i environments this is an unacceptable level of access. Alliance provides you with the ability to restrict a user to a named list of libraries. For example, you can create a library named **DOWNLOAD** where FTP users can download files, and a library named **UPLOAD** where you allow users to upload files to the IBM i. You can then create a library list that includes these two libraries. FTP users who are restricted to this library list will only be able to access and display the contents of these two libraries. All other requests will be rejected.

Controlling user access to files

In addition to library level security Alliance gives you the ability to register a file in any library and define which users can access the file. This provides a more granular approach to file security and lets you give access to specific files in your application libraries.

When you give access to a file in a library you define what functions a user can perform on the file. These function include upload, download, rename, and delete of the file. You can give these authorities selectively to individual users.

Controlling user access to remote commands

IBM i FTP provides a means of accessing system and user commands through the FTP subcommand **RCMD** . You can execute any non-interactive command on the IBM i with this facility. For example, you can execute the Clear Library (**CLRLIB**) command in FTP by entering this on the FTP command line:

```
quote rcmd clrlib ordprd
```

The **ORDPRD** library would be cleared if you have authority to do this on the library. Many IBM i applications depend on menu control to protect production environments, and this operation would be processed. This can represent a clear danger to your IBM i applications.

With Alliance you can control access to the RCMD command and either prevent any use of the command or restrict a user to a list of commands that you define. For power users you can also authorize unrestricted use of the **RCMD** facility.

Implementing Anonymous access

With Alliance you can implement secure access to the IBM i for anonymous users and control access to libraries and IBM i functions. You can control the source IP address, libraries the user can access, and restrict access to FTP and system commands. In addition, Alliance will record all anonymous user access in the history file, including the password the user entered which is, by convention, the user's e-mail address.

Intruder Alert

In addition to providing you with the above access controls, Alliance also includes the ability to monitor for break in attempts and take proactive measures to protect your IBM i. These capabilities are called Intruder Alert. You can set warning and alert thresholds for invalid

access or unauthorized FTP command usage and Alliance will warn you when these thresholds are exceeded. In addition, if your alert threshold is exceeded you can instruct Alliance to terminate FTP support on the IBM i. By terminating FTP support you can eliminate the threat of break in by unauthorized users. Alliance can warn you of potential security threats via Electronic Mail, messages to your terminal or user message queue, and notifications to a user data queue.

Alliance FTP Audit Trails and History

Audit trails

Alliance keeps a history file of all logon and FTP requests. You can view a list of logon attempts that have been accepted and rejected as well as all FTP requests such as downloads, uploads, file lists, system commands, etc. The audit trail is accumulated in a history file named **`AFHIST`** that you can query with IBM's query product or many third party query products.

The history file gives you a way of periodically reviewing use of FTP on your system. You can identify potentially inappropriate uses of FTP or attempts to gain unwelcome access to your IBM i. All of the information that IBM makes available through the security exit points for FTP are logged in the history file.

If you have implemented Anonymous access to FTP (see below), the history file will have a record of the password the user entered. By convention this is normally the e-mail address of the user, but there is no validation of the address. However, if you are making extensive use of Anonymous access you can list and use the e-mail addresses for follow- up activities.

Application logs

To support problem determination and support implementation of Alliance services, you can activate internal logging of the Alliance logon and request validation programs. When logging is active detailed information is written to the log file **`AFLOGA`** . You can view this log with the option to Display Application Log on the main menu, or using any file display or query tool. The application log displays all input parameters to the Alliance control programs and all relevant Alliance options you have configured. If a logon or request is rejected by Alliance the reason for the rejection will be in the application log.

You can also see a summary of the reason for a rejected logon or command in the history file.

If you report a problem with Alliance, your software vendor may ask you to activate the application log and print the results. The log is activated with the option to Configure FTP Security on the main menu.

Common Sense Cautions

If you are not familiar with Internet security precautions, you should familiarize yourself with the normal steps for securing a computer system attached to the Internet before you proceed. Alliance, and similar products, are only one part of a strategy to secure your systems from

unwelcome access. A complete strategy includes many elements, including Firewall systems, proxy servers, network isolation, operational procedures, and other security controls. You should view Alliance as only one part of an overall strategy for security on your IBM i.

Related Publications

You may find these related publications helpful:

TCP/IP Configuration and Reference AS/400 Advanced Series SC41-3420

TCP/IP Fastpath Setup AS/400 Advanced Series SC41-3430

Tips and Tools for Securing Your IBM i AS/400 Advanced Series SC41-3300

FTP Applications and Web Browsers

Overview

To access the IBM i FTP server you will need an FTP application on your client system. If the client system is a Windows PC (3.1, Windows For Workgroups, 95, or NT) or Macintosh, there are several FTP applications you can use. You can also use a Web browser such as Netscape 3.0 or Microsoft Internet Explorer 3.0, or another Web browser that supports FTP. If the client system is an IBM i, you can use the FTP command to start a session with the remote IBM i.

You should note that Web browsers are more difficult to use than FTP applications. The browser will attempt to log on as user Anonymous and will perform several functions to set up the environment. This frequently causes problems with the IBM i. See the notes below on Netscape and Microsoft browsers if you are going to allow FTP access from browsers. You will have much more luck imbedding a URL in a Web page for download than giving Web users the ability to browse your library on the IBM i.

Windows applications

There are several versions of Windows client programs available. If you are using Windows NT Sever, you can find FTP client software for both 16-bit and 32-bit versions of Windows. Windows For Workgroups 3.11 will install an FTP client software application if you install Microsoft TCP/IP. Other vendors also provide FTP client software. Windows95 and Windows NT Workstation both provide FTP applications you can use to connect to the IBM i.

You can also find shareware or low cost solutions. For a shareware FTP application go the www.shareware.com with your browser and search for “ftp.” You will find **`TFTP16.ZIP`** and **`WS_FTP.ZIP`** available for download. For an inexpensive version of Windows FTP try the web site www.tucows.com. The CuteFTP application works well in the Windows environment.

Netscape considerations

Web browsers can present several problems in downloading files from the IBM i. But you can usually resolve these problems with patience and testing. One limitation in Netscape is that Netscape will automatically set the transfer mode to Binary Image. This means that your files

will be downloaded without **ASCII** to **EBCDIC** conversion. If the files are **EBCDIC** they will be garbage on PC or other **ASCII** systems. You must first convert the files to **ASCII** on your IBM i. This is not a problem with the Windows FTP programs mentioned above.

By default, Netscape will attempt to logon as the Anonymous user and will send your e- mail address as the password. It then attempts to set the directory to the root (**/**). This causes a problem for the IBM i and it will normally reject this attempt. To get a list of files in a library, you can enter the URL:

```
ftp://acme.com/qsys.lib/download.lib/
```

Note the use of the expanded file system syntax. You can think of the library **DOWNLOAD** as being a subdirectory of library **QSYS** . To actually download a file you can enter a URL directly for the file:

```
ftp://acme.com/qsys.lib/download.lib/orders.file/myord.mbr
```

This will launch the download of the member **MYORD** in file **ORDERS** in library **DOWNLOAD** .

If your IBM i does not have a DNS entry, you can enter the actual IP address of the IBM i:

```
ftp://198.10.130.5/qsys.lib/download.lib/
```

If you want to override the user and password for the FTP connection in your browser, you can do it on the URL like this:

```
ftp://smith:pass@acme.com/qsys.lib/download.lib/
```

where “smith” is the user profile on the IBM i and “pass” is the password.

Microsoft Internet Explorer considerations

The Internet Explorer application is similar to the Netscape application above, but adds one more complexity to the equation. IE sets the name format to a text version. You can still display the contents of a library with a URL like this:

```
ftp://acme.com/qsys.lib/download.lib/
```

but you will not be able to select a file for download. You will, however, be able to specify the file name on the URL and initiate file transfer:

```
ftp://acme.com/qsys.lib/download.lib/test01.file/test01.mbr
```

IE also allows for the same user ID and password overrides on the URL as Netscape (see above).

Before You Begin

IBM i operating system release level

Before you begin you must verify that your IBM i is at the proper release level. The FTP exit points were added with release V3R2 and V3R7. If you have either of these releases of the IBM i operating system, or a later release, you can use Alliance for FTP security. If you have an earlier release you must first upgrade to V3R2 (or V3R7) or later.

Configuring TCP/IP

If you have not configured the TCP/IP environment on the IBM i, you must do this before you use Alliance FTP security. You must define an IP address (or addresses, for a multi-homed system), provide routing information, and a host and domain name for the IBM i. You should review the configuration guide for TCP/IP before starting. Use the Configure TCP/IP (**`CFGTCP`**) command to define the TCP/IP environment.

Starting TCP/IP

Once you have configured TCP/IP and FTP on the IBM i you must start the FTP servers before FTP will be active. You can use the STRTCPSVR command to start FTP. To start all TCP/IP services you have configured, you can use this command:

```
strtcpsvr *all
```

Or to just start FTP you can enter this command:

```
strtcpsvr *ftp
```

Also, you may want to start TCP/IP services in your IPL start up program. You can determine which program is being run at IPL by displaying the **`QSTRUPPGM`** system value:

```
dspsysval qstruppgm
```

Internet considerations

If you are going to connect your IBM i to the Internet, you should thoroughly review the appropriate security measures for your Internet connection before starting. The advice of an experienced network provider can be invaluable in avoiding security pitfalls with your connection to the Internet.

Installation

Installing for the first time

You may have received Alliance software product in one of several ways:

- On tape media from your software vendor.
- Downloaded from the Internet.
- As a binary save file on PC disk.

In all cases you will restore the product using the Restore Library (**`RSTLIB`**) command.

If you receive the product on tape media, mount the tape on the appropriate device and restore the library from tape:

```
rstlib savlib(allftp130) dev(tap01)
```

If you received the product on PC disk or downloaded it from the Internet, use these procedures:

1. Create a save file on the IBM i: crtsavf qgpl/allftp130 [***Enter***].
2. Start an FTP session with the IBM i from your PC.
3. Change to binary transfer mode: binary [***Enter***].
4. Transfer the file to the IBM i: put a:\allftp.sav qgpl/allftp130 [***Enter***].
5. Close the FTP session and sign on to the IBM i using the **`QSECOFR`** profile. You can now restore the library from the save file:

```
rstlib savlib(allftp130) dev(*savf) savf(qgpl/allftp130)
```

Note: You must restore the product to the library named **`ALLFTP130`** . You cannot change the library to another name.

Installing an upgrade

If you are installing Alliance as an upgrade to a previous release, use these procedures first:

1. End the TCP/IP FTP application: **`endtcpsvr *ftp`** [***Enter***].
2. Save the library **`ALLFTP130`** to backup tape.
3. Rename library **`ALLFTP130`** to a save name.
4. Restore the new release from tape (see above).

5. Run the release update option from the Installation menu.
6. Restart the FTP application: **`strtcpsvr *ftp`** [***Enter***].

Trial and activation codes

If you are installing an upgrade or new release your trial or activation code will be restored when you run the update procedure. If you are installing Alliance for the first time, you will need a trial code from your software vendor.

Alliance library security

The owner of the Alliance library **`ALLFTP130`** and all programs and objects in the library is **`QPGMR`** . The public (**`*PUBLIC`**) has all rights to the library and objects in the library. You should secure the library using normal IBM i security commands.

Planning Your Implementation

Configuring Alliance FTP security is easy, but you will have to make some decisions first about what controls you want and how to implement them. You should decide the following items before starting:

Separate libraries for FTP access

You should determine your policies on allowing FTP user access to libraries on the IBM i before you start. You will probably want to create one or more libraries on the IBM i and restrict FTP user access to these libraries and prohibit access to production libraries. This means you will need to copy files to the new FTP libraries to make files available for download. However you decide to control access to IBM i libraries, be sure to decide this now and have a plan for making files available.

Restricting access to libraries

You should determine your policy on letting FTP users access production libraries on the IBM i. Even for power users you will probably want to allow access only to certain libraries. Determine these policies now so that you can develop library list definitions later and limit FTP users to the appropriate set of libraries.

Anonymous user access to IBM i

The Anonymous user is a special case in FTP access to a host. The intent of Anonymous access is to enable casual users the ability to download files that you make publicly available. By convention in FTP you do not require the Anonymous user to provide a user ID and password that is known to the system. The user will enter “anonymous” as the user ID and their e-mail address as the password. They will then be associated with a download library and can download the files in that library. Generally, the only functions allowed to an Anonymous user is the ability to list the files in your library and download one or more files. No other

function is generally permitted an Anonymous user. Before you begin you should decide if you are going to implement support for Anonymous access. If you decide not to allow Anonymous access you can change this policy later.

Unregistered user access to IBM i

An unregistered FTP user is a user that is not known to Alliance FTP security. As the system administrator you have the option of allowing unregistered users access to the IBM i. They will still need a valid user profile and password, but Alliance will not enforce library or command restrictions on these users. For the best security control it is recommended that you not allow unregistered users. However, in some cases this will be appropriate. You should determine you policy on unregistered users before you start.

Allow access to remote commands

FTP provides a means of executing operating system commands on the host system. On the IBM i, FTP implements the Remote Command subcommand (`RCMD`). With this facility it is possible for client users to execute commands and call programs. It will probably be inappropriate for normal FTP users to execute remote commands and you will want to prohibit this FTP subcommand. Before you start you should determine if there is any legitimate need for the remote command facility and plan on defining which commands will be allowed. If there is no need for using remote commands you can simply deny this capability to any FTP user.

Restricting FTP access to IP addresses

Alliance will allow you to restrict global access to the IBM i to a set of client IP addresses or partial addresses. This can provide an additional level of security based on the user's actual system IP address. If you can use this facility you should plan on defining these addresses. If you are allowing Internet users to access the IBM i, you will probably not be able to implement this control, as the users will have a very large variety of IP addresses. The same user may even have different IP addresses at different times if their Internet Service Provider dynamically assigns them an IP address. This is common for dial connections to the Internet.

If you are implementing FTP access on your own internal network you will probably be able to implement controls based on the IP address. Your network administrator will be able to give you a list of IP addresses for FTP users.

Alliance will also allow you to control access based on just a portion of the IP address of clients. You can use your TCP/IP network address or network address plus a portion of the IP address if you want. For example, if you have a corporate network that uses the network address of `231.*.*.*` , you can restrict FTP access just to those users within your corporate network by specifying `231.`

Implementing client IP address control gives you one more level of protection against unwelcome FTP access. Before you begin you should determine your policy on IP address control.

Restricting user access based on IP address

In addition to being able to restrict FTP access on a global basis you can restrict individual user access based on IP address. On most networks each PC or Workstation that accesses FTP support on the IBM i will have a unique IP address. If this is the case you should consider restricting the user to FTP access based on their specific IP address. In some networks, however, IP addresses are assigned dynamically to users on the network. In this case you will not be able to implement IP address control on a per-user basis. Your network administrator can tell you if each user has a pre-determined IP address when they start an FTP session.

Planning For Intruder Alert

The Intruder Alert feature of Alliance will let you set warning and alert levels for FTP access on the IBM i. When warning levels are exceeded Alliance will send you a terminal message or electronic mail message indicating the possibility of a break-in. When the alert level is exceeded Alliance will send a message to this effect. You also have the ability to instruct Alliance to terminate FTP support on the IBM i if the alert level is exceeded. If you enable this option Alliance will terminate FTP support and no further access will be allowed until FTP is restarted. Before you begin you should identify the message queues and electronic mail addresses you would like to notify for warning and alert conditions. Additionally, if you enable the option to terminate FTP support you should develop a plan for analyzing the source of the break in attempt and restarting FTP support on the IBM i.

Getting Started

Overview

The following sections provide a step-by-step procedure for setting up Alliance FTP security. Some of the steps may not be necessary depending on your objectives outlined above.

Creating libraries for FTP

If you want to restrict your FTP users to separate libraries on the IBM i, you should create those libraries now. For example, you might want to create a library called **`DOWNLOAD`** to contain any files you want to make available as general downloads. You can create the library with this command:

```
crtlib download text('FTP download files') aut(*all)
```

If you have several different types of users and you want to keep the files separate, create other libraries to hold files for each group.

Creating the anonymous user ID

If you are going to allow anonymous user access to the IBM i, you should create a user profile for that user. The user profile can have any name you want. Assign the user profile a password and minimal authorities. As an example:

```
crtusrprf usrprf(anonymous) pass(m187f94) usrcls(*user) text('FTP a
```

Use your own password and record the password and profile name for use when you define FTP users later on.

If you want to give access to other users who do not have a user profile on the IBM i, create those user profiles now.

Defining a library list

You will probably want to restrict your FTP users to a specific set of libraries on the IBM i. For example, you might want to restrict Anonymous FTP users to the **DOWNLOAD** library you created above. You should define these library lists now and use the lists when you define FTP users below. Use the Work With Library Lists option from the main menu and use ***F6*** to create the library lists. For example, you could create a list named **ANONYMOUS** and add only one library to the list: **DOWNLOAD** .

Create any other library lists you want to use with FTP users now.

You do not have to restrict an FTP user to a library list. For power users you can disable library list control and allow the user to access any library to which they are authorized by their user profile.

Define FTP remote commands

You may not have any users you want to give the FTP Remote Command capability to. This would normally be the case. After all, if a user has a need to execute commands on the IBM i you will probably give them a terminal emulation program or other normal terminal access to the IBM i for this purpose. However, you may have applications that will execute remote commands, or you may want to give some power users the ability to run IBM i commands. If so, create a command list now for these users. Use the **Work With Remote Commands** option on the main menu to create a command list. The command list is referenced when you create FTP command lists in the next step.

You should probably never enable remote commands for an anonymous user.

Defining FTP command lists

You will probably want to restrict the FTP commands that a user can access while in an FTP session. For example, there will probably never be a time when you will want an FTP user to use the **RMDIR** command to delete a library. And there will probably be users who you will not want to use the **DEL** or **RM** command to delete files. Use the **Work With FTP Commands** menu option to create an FTP command list.

For the anonymous user you may only want to allow the ability to list files and download (send) files. You could create a list with the name **ANONYMOUS** and enable only these two options. The options to create a library, delete a library, delete a file, execute a remote command, and other options should not be allowed.

For other groups of users you should create appropriate command lists for the abilities you want to give them.

Define the Anonymous FTP user

If you want to allow Anonymous FTP access to the IBM i, you should create this user definition now. Create the user definition with the Work With FTP Users option on the main menu. Be sure to reference the library list and FTP command list you created for anonymous access above. Be sure to select the option to override the password to the password you gave to the anonymous user profile.

Define the unregistered FTP user

If you are going to allow unregistered user access to FTP, and if you are going to override these unregistered users to a specific user ID, define that user now. Use the **`Work With FTP Users`** option from the main menu to do this. Be sure to specify appropriate library list and FTP command list definitions.

Defining FTP users

You should now define other FTP users to Alliance. The information you created above will be input to this process. Use the Work With FTP users option on the main menu to define users. Define the users with the same name as their user profile on the system. As you define each user be sure to provide an appropriate library list and FTP command list definition for the user. Additionally, if the user will have a permanent IP address and you want to restrict the user to that address, provide that information now.

Creating an IP address table

If your FTP users will be accessing the IBM i from one or more known networks and will have predictable IP addresses or network addresses, you should create an IP address table with this information now. If you will be allowing Internet access to the IBM i you may not be able to provide IP address restrictions. However, if your Internet users first pass through a firewall to the IBM i, the firewall may be assigning known IP addresses as a part of the proxy process. Consult with your network administrator if this is the case. If you can use IP address control, you should do so as it provides greater security for your IBM i. Create the IP address list with the **`Work With IP Addresses`** option on the main menu.

Configuring FTP security

After defining all of your FTP users on the system, you should configure global options for Alliance FTP security. Use the Configure FTP Security option on the main menu. Be sure to provide the options for anonymous and unregistered users at this time. Also, if your FTP users will be accessing the IBM i from a known network, you should provide an IP address table definition now.

Activating Alliance FTP security

The last step is to activate Alliance FTP security by registering Alliance programs as exit programs to FTP logon and request exit points. You can do this using the Enable FTP Logon Security and Enable FTP Request Security options on the main menu. Alliance will not be functional until you take this step. After using these menu options it will be necessary to stop the FTP server, if it is active, and start it again. To end FTP server functions you can use this command:

```
endtcpsvr *ftp
```

It will take a few moments for the FTP server jobs to terminate. When they are finished you can start FTP again with this command:

```
strtcpsvr *ftp
```

You should now check each of the users you defined to the system to verify that the policies you developed for access to libraries, FTP commands, and system commands are now in effect. You can use the Display History Log option on the main menu to view the results of each logon request and each command request you make.

Checklist

The following checklist is provided to assist in the configuration and startup process. Some steps may not apply to you, so just skip those sections. See the sections above for more information on the following steps.

STEP	COMPLETE	DESCRIPTION
1		Review IBM i operating system requirements before starting.
2		Review the TCP/IP configuration and be sure that the FTP server application is configured. You can work with the TCP/IP and FTP configuration using the `CFGTCP` command.
3		Restore library `ALLFTP130` from product media.
4		Enter your temporary or permanent activation code from your softv vendor using the appropriate menu option. To display the menu: `_addlible allftp130` [***Enter***] to `almain` [***Enter***]_
5		Create the libraries you will use for FTP. For example, you might create a library named `DOWNLOAD` for all files you will make availa to FTP users: `_crtlib download aut(*all) text('FTP download libra` [***Enter***]_

STEP	COMPLETE	DESCRIPTION
6		Create a library list definition for your FTP libraries. Alliance uses library list definitions to restrict access to libraries. Use the **`Work With Library Lists`** option from the menu. You will wan include any libraries you want to make available to FTP users.
7		Are you going to implement Anonymous FTP support? If so, creat user profile for the Anonymous user. It can be named anything you want. For example: **`_crtusrprf anonymous pass() usrcls(*user)`** [***Enter***]_ Use your own password or specify **`*NONE`** .
8		Create an FTP command list for your FTP users. Use the option o menu to **`Work With FTP Commands`** . Add a list that includes just List Files, Download Files, and Change Directory commands.
9		Create Alliance FTP users who will use FTP. Use the **`Work With FTP Users`** options on the main menu. Add the user the Anonymous FTP account if you are going to implement Anonymous FTP support, and add other users who will use FTP.
10		If you are going to restrict FTP users based on an IP address or network address, create an IP address table. Use the **`Work With IP Addresses`** option on the main menu. Include all addresses or network addresses that will be authorized to use FTI
11		If you are going to use Intruder Alert you will need to identify up to three message queues and/or three electronic mail addresses to receive alert messages. This information is provided in the next st when you configure FTP security.
12		Configure Alliance FTP security. Use the option to Configure FTP Security on the Alliance main menu.
13		Activate Alliance FTP logon and request security. Use the Enable Logon Security option and the Enable FTP Request Security optio on the main menu.
14		If the FTP server jobs are active, you must stop them now. Use th **`ENDTCPSVR`** command: **`_endtcpsvr *ftp_`**
15		Start the FTP server jobs. Use the **`STRTCPSVR`** command: **`_strtcpsvr *ftp_`**
16		Use an FTP client application to download files from the IBM i. Be your security policies are working properly.

STEP	COMPLETE	DESCRIPTION
17		Review history audit logs on the IBM i. Use the `View History L` option from the main menu.

On Line Help

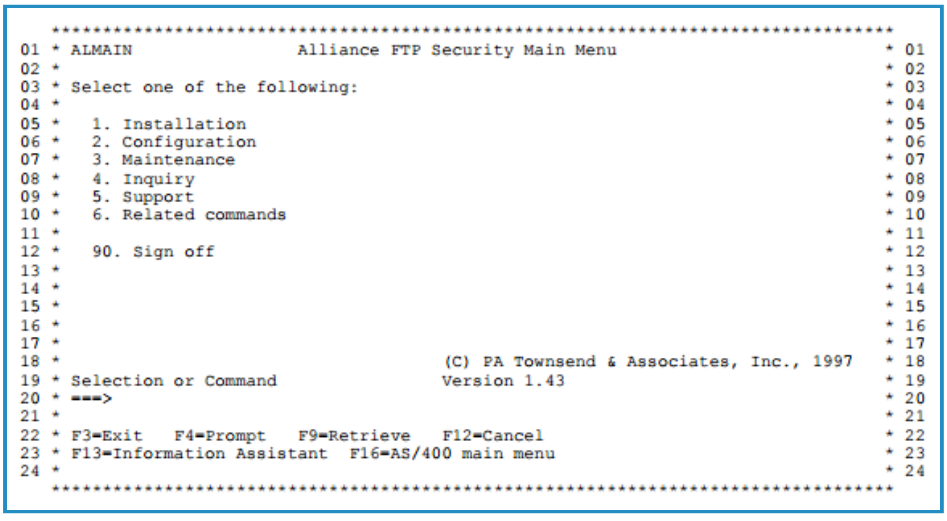
Every panel in Alliance has on-line help available. Press the ***Help*** or ***F1*** key on your keyboard to access help. You will be able to review a description of the panel's function, each field's function and use, and a list of active function keys and what they do.

Main Menu

The main menu for Alliance FTP security is **`ALMAIN`** . You can access this menu by placing the library **`ALLFTP130`** in your library list and using the GO command:

```
addlible allftp130       [**_Enter_**]
go almain                          [**_Enter_**]
```

The following menu is displayed. All of the Alliance programs are available from this menu and sub-menus.

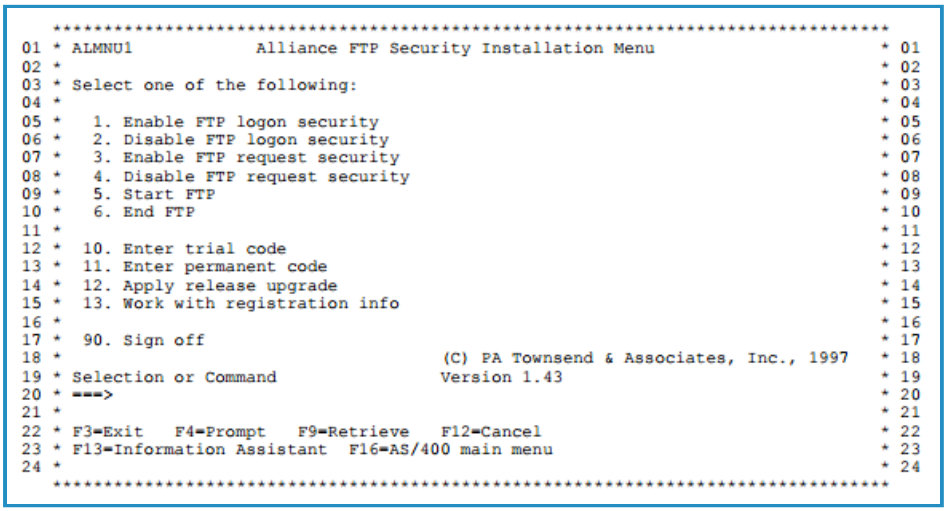


Selection or Command

Enter a menu option for the function you want to use. Or, enter **`90`** to signoff. You can also use function keys to use the IBM i Information Assistant and other functions.

Installation

You can access all of the installation functions from the main menu by selecting option `1` from the main menu. The following menu is displayed.



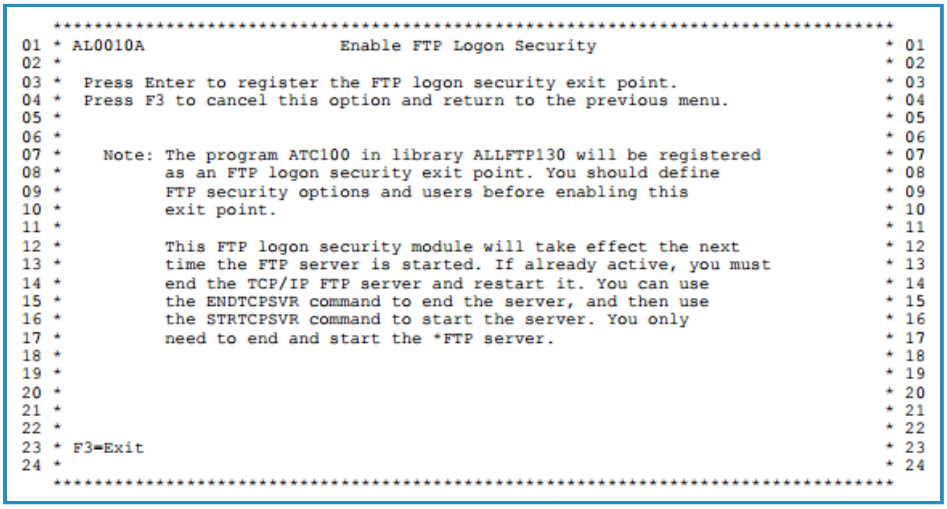
Select an option from the menu to begin work.

Enable FTP Logon Security

Alliance FTP Logon security will not be active until you use this module. This module adds the exit point information for the FTP application. Until you add the exit point information, the IBM FTP application will not call Alliance for security and history functions.

FTP logon security is performed once each time a user connects to the IBM i and enters a user ID and password. If a user issues the FTP command to re-logon (the **`REIN`**) command, the logon security module is called again for the new user ID.

Please note that this option only enables FTP logon security. There is a separate option to enable FTP request security.



There are no input fields on this panel. Press ***Enter*** to add the exit point information for FTP logon security. Press ***F3*** to exit this panel.

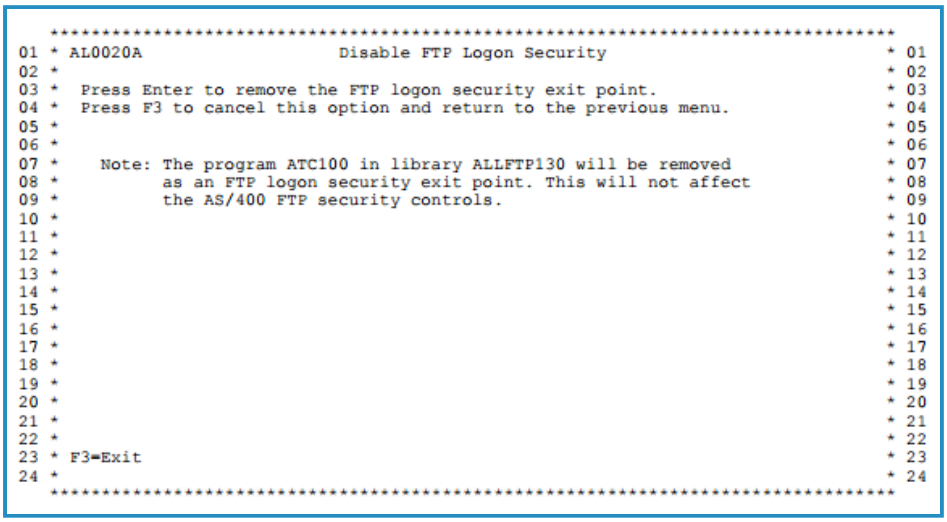
It might be helpful to note that there are two ways to remove Alliance from FTP security once it has been installed with this option: 1) you can use the menu option to disable FTP logon security, or you can use the Configure FTP security option and select to not perform FTP logon

security. This latter method is very easy and avoids the need to stop and restart the FTP application. The Alliance application is still invoked by the IBM FTP application, but no validation is performed. In effect the native FTP environment takes effect again.

Disable FTP Logon Security

Use this option to disable Alliance FTP logon security on your IBM i. This module will remove the exit points in the FTP application that call the Alliance programs for logon security monitoring. Alliance will no longer be invoked each time a user signs on to FTP on the IBM i.

Please note that there is a separate menu option for disabling Alliance FTP request security. This option only disables logon security.



There are no input fields on this panel. Press ***Enter*** to process this request, or ***F3*** to quit without disabling FTP logon security.

Enable FTP Request Security

The Alliance FTP Request security module processes every action the FTP user attempts once they are signed on to the IBM i. In order for the Alliance request security module to be activated you must use this option to install the exit point for Alliance. Press Enter to install the exit point.

Please note that this option only activates FTP request security. There is a separate option to activate FTP logon security.

```
AL0040A                 Enable FTP Request Security

 Press Enter to register the FTP request security exit point.
 Press F3 to cancel this option and return to the previous menu.


   Note: The program ATC200 in library ALLFTP130 will be registered
         as an FTP logon security exit point. You should define
         FTP security options and users before enabling this
         exit point.

         This FTP logon security module will take effect the next
         time the FTP server is started. If already active, you must
         end the TCP/IP FTP server and restart it. You can use
         the ENDTCPSVR command to end the server, and then use
         the STRTCPSVR command to start the server. You only
         need to end and start the *FTP server.


F3=Exit
```

There are no input fields on this panel. Press ***Enter*** to install the Alliance exit point program for the FTP Request security option. Press ***F3*** to leave this panel and not install the Alliance exit point program.

Disable FTP Request Security

You can remove Alliance FTP Request security from the system with this option. When you use this option the Alliance exit point for FTP Request validation is removed. Alliance will no longer be called for request validation and security.

If you want to temporarily disable Alliance FTP request security, you can use the Configure FTP Security option from the menu to change the status of Alliance FTP request security processing. Alliance will still be called by the IBM FTP application, but will perform no further request validation and security services until re-activated. After using this option you must use the option to Enable FTP Request Security in order to re- activate Alliance processing.

```
AL0050A                 Disable FTP Request Security

Press Enter to remove the FTP request validation security exit point.
Press F3 to cancel this option and return to the previous menu.


   Note: The program ATC200 in library ALLFTP130 will be removed
         as an FTP logon security exit point. This will not affect
         the normal AS/400 FTP security controls.


F3=Exit
```

There are no input fields on this panel. Press ***Enter*** to remove the exit point programs for Alliance FTP request security. Press ***F3*** to leave this panel without removing the exit point programs.

Start FTP

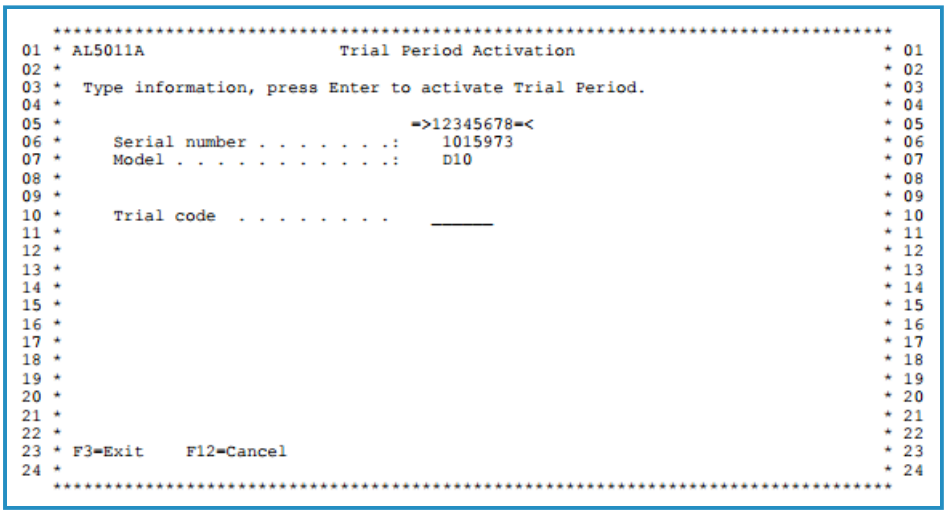
This option on the menu initiates the FTP server using the IBM i Start TCP/IP Server (**STRTCPSVR**) command. It is provided on this menu for convenience. You will need to start the FTP server before any user is allowed access to the IBM i, and after Intruder Alert has ended FTP access if you have this option enabled.

End FTP

This option terminates the FTP server functions of the IBM i using the End TCP/IP Server (**ENDTCPSVR**) command. It is provided on this menu for convenience. You will need to end and then restart FTP support after you have completed configuring Alliance FTP.

Enter Trial Code

To use Alliance FTP security you must have a trial code or permanent code. A trial code was provided with your shipment and will be active for approximately 30 days. Use this panel to enter the trial code.

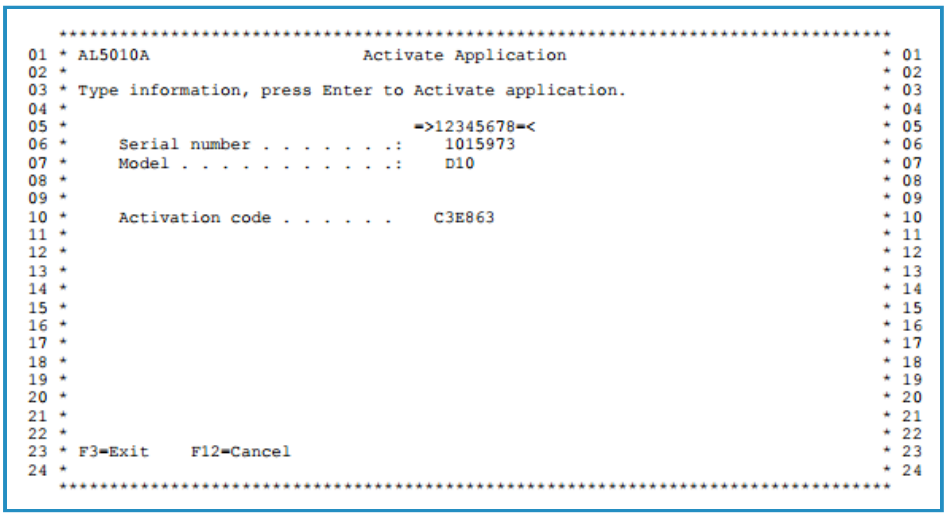


Trial code

Enter the trial code provided by your software vendor.

Enter Permanent Activation

After you have licensed Alliance for your IBM i you will receive a permanent activation code. Enter the permanent code in this panel.

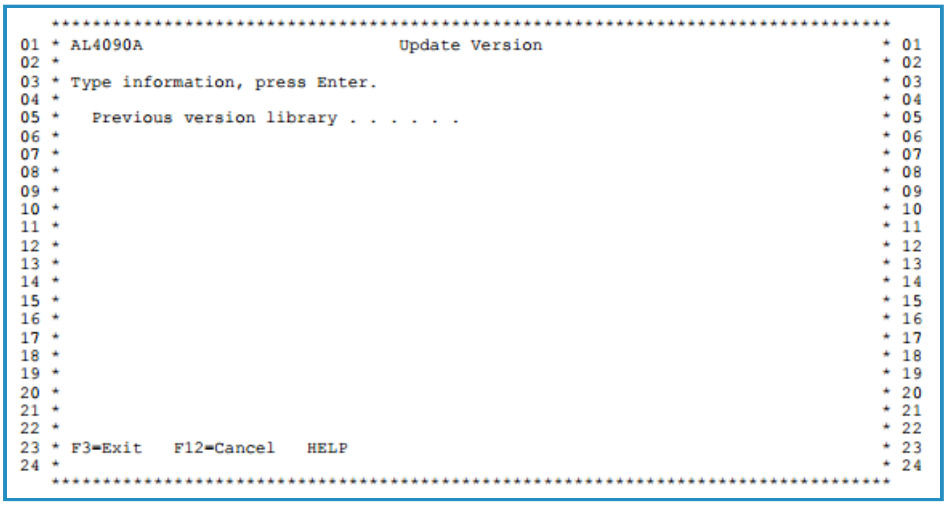


Activation code

Enter the permanent activation code provided by your software vendor.

Apply Release Upgrade

If you receive a release upgrade from your software vendor you will use this option to re-establish your FTP options, user lists, and other configurations you developed on the earlier release. Please be sure to read the installation cover letter that accompanies the release upgrade before using this option. When you select this option the following panel is displayed.



Previous version library

Enter the name of the library that contains the previous release.

Configuration

When you select option **2** from the main Alliance menu the following menu is displayed. You can configure all of the Alliance FTP options and tables from this menu. Select an option to begin work.

```
                                                                                
ALMNU2            Alliance FTP Security Configuration Menu                      
                                                                                
Select one of the following:                                                    
                                                                                
   1. Configure FTP security                                                    
   2. Work with FTP users                                                       
   3. Work with FTP commands                                                    
   4. Work with FTP remote commands                                             
   5. Work with IP addresses                                                    
   6. Work with library security                                                
   7. Work with file security                                                   
                                                                                
  90. Sign off                                                                  
                                                                                
                                                                                
                                                                                
                                                                                
                                     (C) PA Townsend & Associates, Inc., 1997   
Selection or Command                 Version 1.43                               
===>                                                                            
                                                                                
F3=Exit    F4=Prompt    F9=Retrieve    F12=Cancel                               
F13=Information Assistant  F16=AS/400 main menu                                 
                                                                                
```

Configure FTP Security

Use this panel to configure global options for Alliance FTP security. You can activate or de-activate logon and request security, disable FTP access to the IBM i, and establish global options related to Anonymous FTP, unregistered FTP users, and available libraries on the IBM i.

```
                                                                                
AL6040A                    Configure FTP Security                               
                                                                                
  Press Enter to continue, or press F3 or F12 to quit.                          
                                                                                
  Enable FTP logon security  . . . . .   1              1=Yes, 2=No             
  Enable FTP request security  . . . .   1              1=Yes, 2=No             
                                                                                
  Allow FTP access . . . . . . . . . .   1              1=Yes, 2=No             
                                                                                
  Allow unregistered users . . . . . .   1              1=Yes, 2=No             
  Override unregistered users  . . . .   2              1=Yes, 2=No             
    User ID  . . . . . . . . . . . . .                  F4=Prompt               
                                                                                
  Allow ANONYMOUS FTP access . . . . .   2              1=Yes, 2=No             
    ANONYMOUS user ID  . . . . . . . .                  F4=Prompt               
                                                                                
  Enable IP address control  . . . . .   1              1=Yes, 2=No             
    IP address table . . . . . . . . .      IPTABLE     F4=Prompt               
                                                                                
  Enable logging . . . . . . . . . . .   1              1=Yes, 2=No             
                                                                    More...     
                                                                                
F3=Exit    HELP                                                                 
                                                                                
```

Enable FTP logon security

Enter **`1`** for **`Yes`** if you want Alliance to provide FTP logon security functions. Enter **`2`** for **`No`** if you do not want to use Alliance FTP logon security. If you have registered Alliance as the exit program for logon security you can temporarily disable Alliance security processing by setting this value to **`2`** . Normal IBM i FTP security will still be in effect. Please note that Alliance FTP logon security will not be in effect, regardless of this setting, unless you take the option to Enable FTP Logon Security from the main menu.

This value gives you a way to quickly enable and disable logon security after registering the Alliance exit point modules.

Enable FTP request security

Enter **1** for **Yes** if you want to want Alliance to provide FTP request security functions. Enter **2** for **No** if you do not want to use Alliance FTP request security. If you have registered Alliance as the exit program for request security you can temporarily disable Alliance security processing by setting this value to **2** . Normal IBM i security will still be in effect. Please note that Alliance FTP request security will not be in effect, regardless of this setting, unless you take the option to Enable FTP Request Security from the main menu. This value gives you a way to quickly enable and disable request security after registering the Alliance exit point modules.

Allow FTP access

Enter **1** for **Yes** to enable FTP access to the IBM i or **2** for **No** to disable FTP access to the IBM i. If you enter **2** the IBM i will accept connection requests for FTP clients, but Alliance will never allow a user to sign on to the IBM i. This option is designed to give you a quick way to enable and disable FTP access without terminating the FTP server.

Allow unregistered users

Enter **1** for **Yes** if you want to allow unregistered users or **2** for **No** if you do not want to allow unregistered users. A registered user is a user you have added to the Alliance FTP users file. You can add users to the Alliance FTP users file by using the Work With FTP Users option on the main menu. If you enter **1** to allow unregistered users, Alliance will allow the user to sign on to the IBM i with their user ID and password entered from the FTP signon. The intent of this option is to give the system administrator better control over FTP access to the IBM i and it is recommended that you not allow unregistered users. If an unwelcome user guesses a user ID and password for the IBM i, their sign on attempt will be rejected if they are not in the Alliance FTP users file.

Override unregistered users

Enter **1** for **Yes** if you want to override an unregistered user to an Alliance FTP user that you have already defined. Enter **2** for **No** if you do not want to override unregistered users to an Alliance user ID. The intent of this option is to allow you to force unregistered users to a known Alliance FTP user name. This could be a name like ANONYMOUS, or any other Alliance FTP user name you specify. By forcing unregistered users to a known user ID, you can control many aspects of their session on the IBM i. It is recommended that you specify **Yes** for this option and provide an Alliance FTP user name in the next field.

Unregistered User ID

If you selected **Yes** for **Override Unregistered Users** , you must enter an Alliance FTP user name for this value. If you specified **No** for **Override Unregistered Users** , you can leave this field blank. If you enter a user ID for this field you must first define the user ID with the **Work With FTP Users** option on the main menu.

Allow Anonymous FTP access

Enter **1** for **Yes** if you want to allow the user ANONYMOUS to sign on to IBM i FTP. Enter **2** for **No** if you do not want to allow access to the user ANONYMOUS. The user ANONYMOUS is a special user that is often implemented on an FTP site. A casual user can enter the User ID of “ANONYMOUS” and their e-mail address for the password, and gain access to the general upload and download library. This option will override the TCP/IP configuration option for Anonymous access. That is, even if you specify with TCP/IP configuration that you want to allow Anonymous access, these users will be rejected if you specify **No** for this value. If you specify **Yes** , you must provide the name of an Alliance FTP user name in the following field.

Anonymous user ID

If you specified **Yes** to allow anonymous FTP access you must specify an Alliance FTP user ID. Use the **Work With FTP Users** option on the main menu to create an Alliance FTP User ID. For the sake of clarity, it is recommended that you create an Alliance FTP user named “ANONYMOUS” and give it the capabilities you want for anonymous users. Then enter that name in this field.

Enable IP address control

Enter **1** for **Yes** if you want to implement IP address control for access to the IBM i FTP application. Enter **2** for **No** if you do not want to implement IP address control for access to the IBM i. IP address control will let you specify the client IP address, or portion of the IP address, for access to the IBM i. This allows you to reject any attempt to log on to the IBM i from a client workstation or PC that is not on a known internal network. If you are connecting the IBM i to the Internet you probably won’t be able to use this option as the users are often assigned an IP address dynamically, and the address will be different each time the user signs on. Also, some network systems, such as Microsoft’s TCP/IP with DHCP, dynamically assign IP addresses. However, even in this case, you can use IP address control and specify the network portion of the address. If you are using a firewall and/or proxy servers for FTP access you may still be able to define IP address control. Consult your network administrator for more details on the assignment of IP addresses to FTP clients. If you can use client IP address control it gives you more security over access to the IBM i.

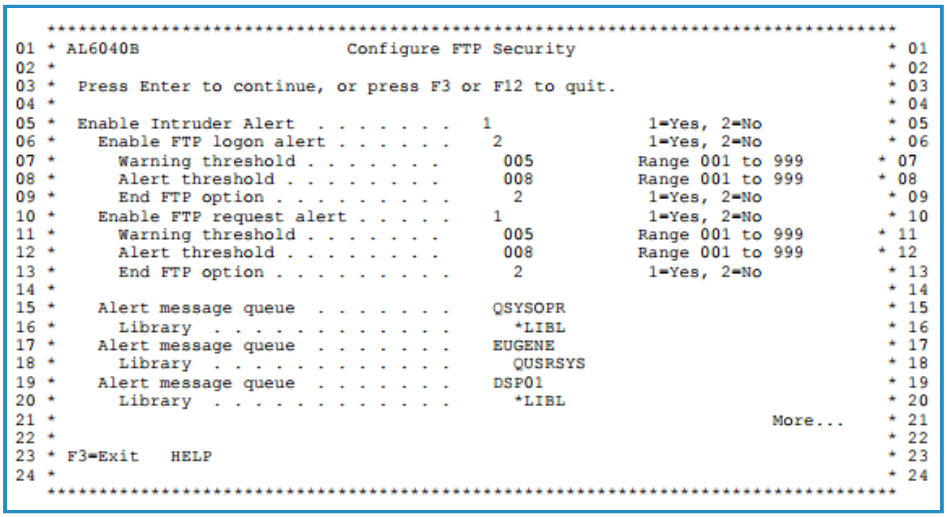
IP address table

If you enabled IP address control you must enter the name of an Alliance IP address control table. Use the **Work With IP Addresses** option from the main menu to create an address table. An IP address table is simply a named list of IP addresses or network addresses.

Enable logging

Enter **1** for **Yes** to enable logging. Enter **2** for **No** to disable logging. When logging is enabled detailed status information is written to the Alliance logging file. The name of this file is **AFLOGA** . You can view log entries with the **Display Application Log** option on the main menu. Logging is designed to assist you in implementing and trouble- shooting Alliance FTP security controls. If you find that a user’s access is not working the way you think it should, you can enable logging and view the results of the user’s session. If a user is being rejected during a signon attempt, or an FTP function is not allowed, the log will have diagnostic information on the reason for the action.

When you press ***Enter*** on the first panel the following panel is displayed for additional information.



Enable Intruder Alert

Enter **`1`** for **`Yes`** if you want to enable the Intruder Alert function of Alliance FTP. If you enable this option Alliance will send warning and alert messages when the access and request error thresholds are exceeded. You can also indicate to Alliance that IBM i FTP support should be ended if the alert threshold is exceeded. Enter **`2`** for **`No`** if you do not want to enable Intruder Alert.

Enable FTP logon alert

Enter **`1`** for **`Yes`** if you want to enable FTP logon alerts. You must have entered **`1`** for **`Yes`** to the Enable Intruder Alert option in order to enable this option. If you enable logon alert Alliance will send warning and alert messages when the access thresholds are exceeded. Each time Alliance denies an FTP user access for any reason the logon reject counter is incremented. When the reject counter exceeds the threshold you define a warning or alert message is sent. Enter **`2`** for **`No`** if you do not want to use FTP logon alerts.

Warning threshold

If you enabled FTP logon alert you must specify a warning threshold. This is the number of sequential rejects upon which Alliance will send a warning message. The valid range is from **`0001`** to **`9999`**.

Alert threshold

If you enabled FTP logon alert you must specify an alert threshold. This is the number of sequential rejects upon which Alliance will send an alert message. The valid range is from **`0001`** to **`9999`**. If you enable the End FTP option in the next parameter this is also the threshold at which FTP services are ended on the IBM i.

End FTP option

Enter **1** for **Yes** if you want Alliance to end IBM i FTP services when the alert threshold is exceeded. This will deny any access to the IBM i from FTP clients until you restart the FTP server. Enter **2** for **No** if you do not want to end FTP services when the alert threshold is exceeded.

Enable FTP request alert

Enter **1** for **Yes** if you want to enable FTP request alerts. You must have entered **1** for **Yes** to the Enable Intruder Alert option in order to enable this option. If you enable request alert Alliance will send warning and alert messages when the FTP request thresholds are exceeded. Each time Alliance denies an FTP user access to an FTP operation for any reason the request reject counter is incremented. When the reject counter exceeds the threshold you define a warning or alert message is sent. Enter **2** for **No** if you do not want to use FTP request alerts.

Warning threshold

If you enabled FTP request alert you must specify a warning threshold. This is the number of sequential rejects upon which Alliance will send a warning message. The valid range is from **0001** to **9999**.

Alert threshold

If you enabled FTP request alert you must specify an alert threshold. This is the number of sequential rejects upon which Alliance will send an alert message. The valid range is from **0001** to **9999**. If you enable the **End FTP** option in the next parameter this is also the threshold at which FTP services are ended on the IBM i.

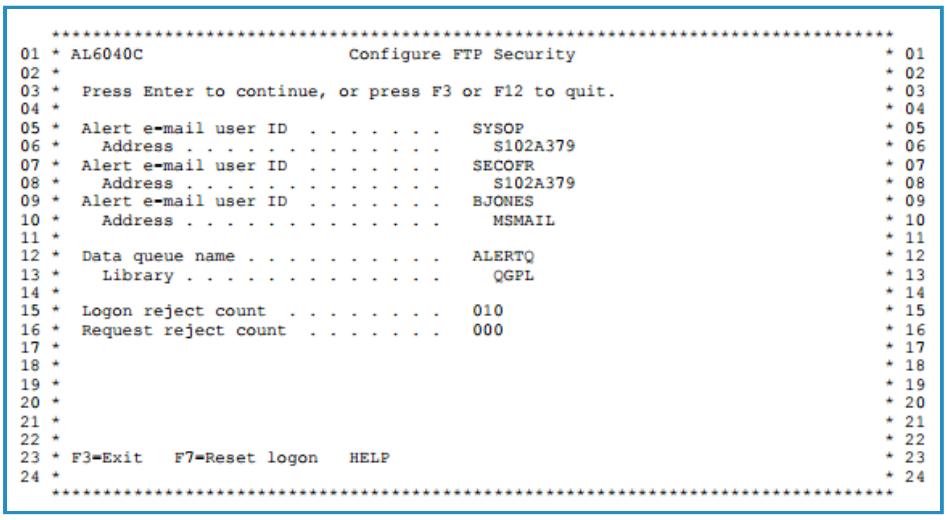
End FTP option

Enter **1** for **Yes** if you want Alliance to end IBM i FTP services when the alert threshold is exceeded. This will deny any access to the IBM i from FTP clients until you restart the FTP server. Enter **2** for **No** if you do not want to end FTP services when the alert threshold is exceeded.

Alert message queue and library

If you have enabled Intruder Alert you can specify up to three message queues to receive warning and alert messages. The message queues can be any valid user, workstation, or application message queue.

When you press ***Enter*** on the second panel the following panel is displayed for additional information.



Alert e-mail user ID and address

If you have enabled Intruder Alert you can specify up to three e-mail addresses to receive warning and alert messages. The e-mail addresses must be any valid SNADS address in your system distribution directory. If you have enabled STMP support in your mail environment, you can send warning and alert messages to Internet recipients.

Data queue name and library

You can also specify the name of a data queue to receive warning and alert messages. This allows you to define a user application to process the warning messages. The user application can take any action you want upon notification of a potential break in.

Logon and Request reject counts

These output only fields show you the number of sequential rejects that have been logged for logon and request functions. You can reset these counts using function key ***F7***.

Work With FTP Users

Use this option to define Alliance FTP users. If you are not allowing unregistered users to access IBM i FTP, you must first add the FTP user with this module. The user name you enter will be the same name as the User Profile. When you define a new Alliance FTP user you will give that user a set of specific authorities. You can also limit the user to a particular IP address to provide an additional level of security.

```
AL7050-A                      Work With FTP Users
Position to . . . .
Type options, press Enter.
  2=Change   4=Delete   6=Print
                                      Allow
Opt User        Description           FTP  Directory
    ANONYMOUS   Anonymous               1  DOWNLOAD
    ROBERTS     Beth Roberts            1  DOWNLOAD
F3=Exit   F5=Refresh   F6=Add   F12=Cancel   F16=Search   F21=Print list
```

To add a new Alliance FTP user, press ***F6***. See the following section for information on adding a new user.

To change information for a user, use option `2`. See the following section for information on changing user information.

To delete an FTP user, use option `4`. You will be prompted to confirm your delete request. Once you delete a user you cannot retrieve the information. To re-establish the user you must add the user again.

You can print the details on a particular user with option `6`. You can print the entire list of users with function key ***F21***.

Add/Change FTP User

When you add or change an FTP user, the following panel will be displayed. Type the requested information and press Enter.

```
AL7051-A                      Change FTP User
User ID . . . . . . . . . . . . . :  ROBERTS
Type information, press Enter.
  Description . . . . . . . . . .    Beth Roberts
  Allow FTP access  . . . . . . .    1                    1=Yes, 2=No
  Force password  . . . . . . . .    2                    1=Yes, 2=No
    Password  . . . . . . . . . .
  Default directory . . . . . . .    DOWNLOAD
  Restrict library access . . . .    1                    1=Yes, 2=No
    Library list name . . . . . .      ROBERTS            F4=Prompt
  Restrict FTP commands . . . . .    1                    1=Yes, 2=No
    Command list  . . . . . . . .      ACME               F4=Prompt
  Restrict IP address . . . . . .    1                    1=Yes, 2=No
    IP address  . . . . . . . . .      157.151.241.
F3=Exit    F12=Previous    HELP
```

User ID

Enter the name of the FTP user. This must be the name of the user profile of the user, or the name “ANONYMOUS.” Enter up to 10 characters with no imbedded blanks or special characters.

Description

Enter a description of the user. This would be the user's name, department, or other identifier.

Allow FTP access

Enter **1** for **Yes** to allow FTP access. Enter **2** for **No** to disallow FTP access. Normally you would specify **Yes** when setting up a new user. You can enter **No** when you want to temporarily disallow the user access to FTP on the IBM i.

Force password

Enter **1** for **Yes** to force the password to be the value you specify on the next field. Enter **2** for **No** to not force the password for this user. Normally you would want to specify **No** for this option. However, when you are specifying a user ID for use by the Anonymous user or the user ID for unregistered users, you will want to force the password to be the password for that user profile.

Password

If you entered **Yes** to force the password for this user, enter the password in this field. This must be the password specified on the user profile.

Default directory

Specify the name of the library you want to be the default directory for this user. This will override normal TCP/IP FTP defaults for the default directory. You can specify different libraries for different users. If you leave this field blank the TCP/IP FTP default library will be used. This is normally library **QGPL** .

Restrict library access

Enter **1** for **Yes** if you want to restrict this user to a list of libraries. Enter **2** for **No** if you do not want to restrict this user's access to libraries. You must first create a named list of libraries if you specify **Yes** .

Library list name

Enter the name of a library list you want to specify. This is a required field if you specified Yes to restrict library access. You can use ***F4*** to prompt for a list of library definitions.

Restrict FTP commands

Enter **1** for **Yes** to restrict FTP commands available for this user. Enter **2** for **No** if you do not want to restrict FTP commands. Normally you would want to restrict a user's access to FTP and IBM i commands and you will want to specify **Yes** for this field. If you specify **Yes**

you must enter the name of a command list in the next field. You can define command lists with the **Work With FTP Commands** option on the main menu.

Command list

If you entered **Yes** to restrict FTP commands you must enter the name of a command list in this field. Use the **Work With FTP Commands** option to define a command list. A command list allows you to control a user's access to both FTP commands (like **DIR**, **LS**, **CD**, etc.) and to IBM i commands via the FTP Remote Command (**RCMD**) interface. It is highly recommended that you specify a commands list for your users and restrict them to functions that are appropriate.

Restrict IP address

Enter **1** for **Yes** if you want to restrict a user to a specific IP address or network address. Enter **2** for **No** if you do not want to restrict a user to an IP address. If you specify **Yes** you can specify a complete network address or portion of a network address in the next field. Consult with your network administrator if you do not know the IP address of a user. Your network may assign IP addresses on demand and a user will not have the same address in different sessions. You can, however, probably restrict access based on the network portion of the address.

IP address

If you specified Yes to restrict a user to an IP address, enter the address in this field. You can use portions of a network address rather than the entire address if you like. For example, you can use:

```
151.157.
```

for this field, and the user will only be allowed access when the IP address starts with this value. Be sure to enter the IP address in dotted format with no leading zeros. For example, this will not work:

```
128.002.005.010
```

But this will work fine:

```
128.2.5.10
```

Work With FTP Commands

Use this option to define named lists of FTP commands that you want to authorize your users to perform. In most cases you will want to prevent your FTP users from creating and deleting directories, but you will want them to be able to download files. When you create a command list definition you are specifying these abilities in a named list. You can then use this named list when you define a new user.

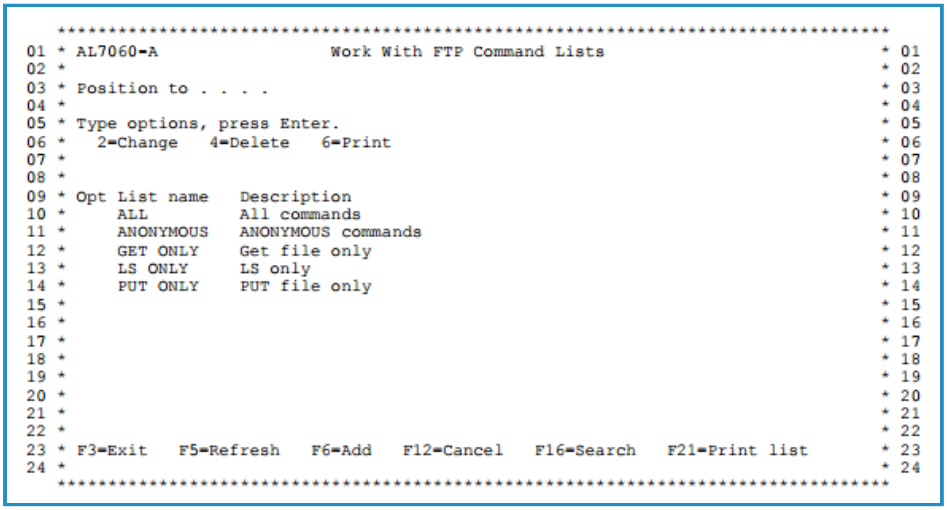
One FTP command is especially important: the FTP Remote Command (**RCMD**). The FTP remote command allows an FTP user to run any non-interactive IBM i operating system command to which they have authority. This can give a user unanticipated access to IBM i system functions. In many IBM i environments, the system administrator controls what functions a user can perform by restricting them to a menu of approved items. The user would never have access to a command line. However, the FTP Remote Command facility lets a user run any command to which they have access via a command line. For example:

```
quote rcmd call invoices
```

would allow a user to access and execute a program named **INVOICES** . And if the user issued this command:

```
quote rcmd pwrdwnsys *immed
```

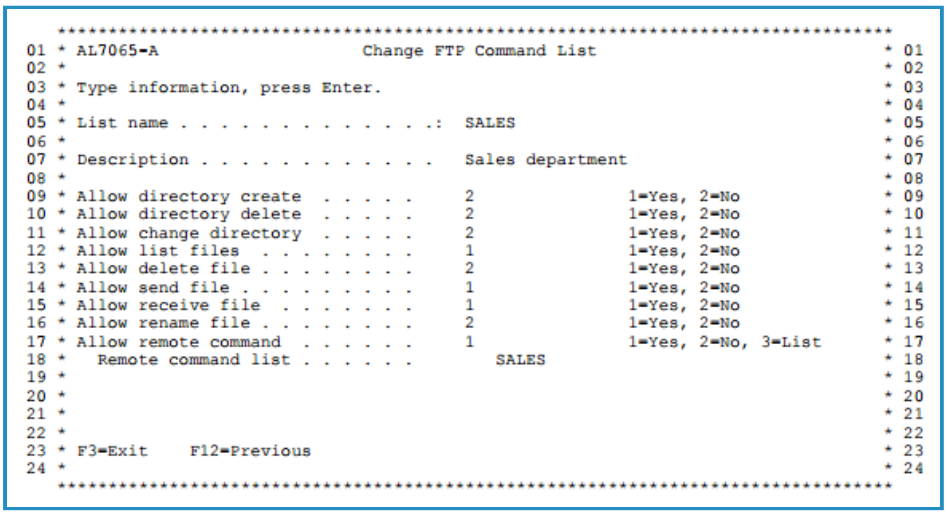
They would power down the IBM i if they had ***USE** authority to the Power Down System (**PWRDWNSYS**) command. You will probably want to prohibit the use of remote commands, or define and use remote command lists for all of your FTP users.



Use ***F6*** to add a new FTP command list. Use option **2** to change an existing command list. Use option **4** to delete a command list. And use option **6** to print a command list. You can also use function key ***F21*** to print a complete list of all command lists.

Add/Change FTP Commands

The following panel is displayed when you create a new list or change an existing command list. Type the information and press Enter to create or update command list information.



List name

Enter a name for this list. The name can be up to 10 characters long and can contain imbedded blanks.

Description

Enter a description for this list. For example, if you are going to create a list that only allows for downloading files you might name the list “DOWNLOAD.”

Allow directory create

Enter **`1`** for **`Yes`** if you want the FTP user to be able to create a directory in an IBM i file system. Enter **`2`** for **`No`** if you do not want the user to be able to create a directory in an IBM i file system. One FTP command for creating a directory is **`MKDIR`** . If you enter the command **`mkdir test01`** you will create a new directory called test01 in the root file system. This is not a library, but is an entirely new directory from the root on the IBM i. If you enter the command **`mkdir /qsys.lib/test01.lib`** you will create a new library named **`TEST01`** . It is recommended that you specify **`No`** for this option.

Allow directory delete

Enter **`1`** for **`Yes`** if you want the FTP user to be able to delete a directory in an IBM i file system. Enter **`2`** for **`No`** if you do not want the user to be able to delete a directory. See the discussion above about IBM i directories. It is recommended that you specify **`No`** for this option.

Allow change directory

Enter **`1`** for **`Yes`** if you want the FTP user to be able to change the default directory. Enter **`2`** for **`No`** if you do not want the FTP user to be able to change the default directory. See the above discussion on IBM i directories. If you allow a user to change the default directory, and you also specify a library list for a user, the user will only be able to change to one of the libraries in the library list.

Allow list files

Enter **1** for **Yes** if you want an FTP user to be able to list files in a directory. Enter **2** for No if you want to disallow the user from listing files. Since you will normally want to allow the user the ability to download files you should set this option to Yes. However, you can restrict this ability by specifying **No** for this field. The FTP commands to list files are **DIR** and **LS** .

Allow delete files

Enter **1** for **Yes** if you want the FTP user to be able to delete files. Enter **2** for **No** if you want to disallow the user from deleting files. If you allow the user to delete files you should consider restricting the user to the default directory or to a library list.

Allow download file

Enter **1** for **Yes** if you want the user to be able to download a file from the IBM i. Enter **2** for **No** if you do not want the user to download a file. If you defined a library list for the user, the user will only be able to download files from the libraries you specify.

Allow upload file

Enter **1** for **Yes** if you want the user to be able to upload a file to the IBM i. Enter **2** for **No** if you do not want the user to upload files. If you defined a library list for the user, the user will only be able to upload files to a library in the list.

Allow rename file

Enter **1** for **Yes** if you want the user to be able to rename files in an IBM i library. Enter **2** for **No** if you do not want the user to be able to rename files. If you defined a library list for the user, only files in the library list can be renamed.

Allow remote command

Enter **1** for **Yes** to allow the FTP user to execute remote commands using the **RCMD** subcommand facility. Enter **2** for **No** to disallow the use of FTP remote commands. Enter **3** for **List** to specify a list of remote commands that the FTP user can access. Normally you will want to enter **2** for **No** to disallow the use of the FTP remote command facility. If the FTP user needs to execute a command or set of commands, you can define a remote command list and authorize the user to the list. Any remote command in the list can be executed by the user.

Remote command list

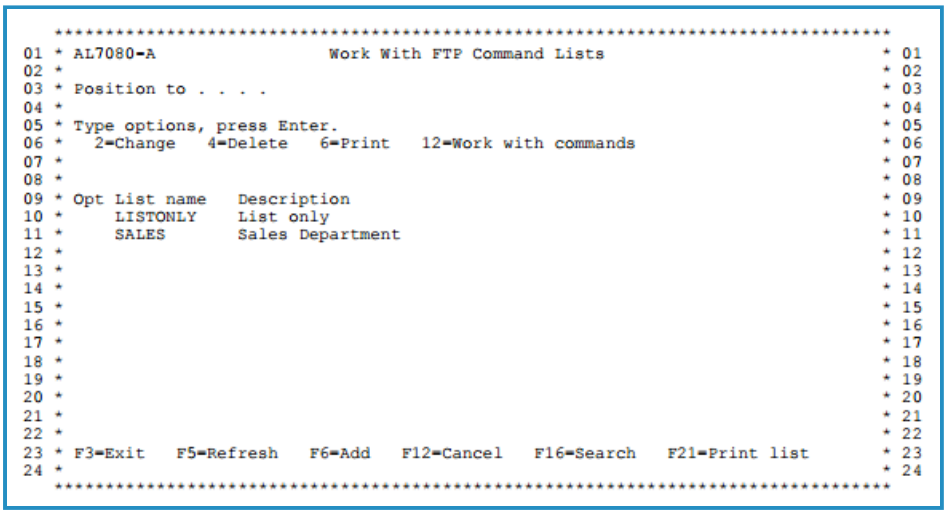
If you specified **3** (**List**) to allow remote commands from a named list, enter the list name in this field. You must first create the list using the **Work With Remote Commands** option on the main menu.

Work With FTP Remote Command Lists

Use this option to create a named list of remote commands you want to make available to FTP users. These are commands that can be executed using the FTP subcommand **RCMD** . Any IBM i command, including user-created commands, can be added to the list of system commands.

You start by creating a named list of commands, and then adding entries to the list. For example, you might create a remote command list named “OPERATIONS” and then add commands like **STRSBS** , **ENDSBS** , **STRWTR** , and other commands an operator might use.

Note that you can create a remote command list and then not add any detail entries to the list. This will have the effect of prohibiting a user from using any remote commands if the list is associated with their FTP commands definition.



Use ***F6*** to create a new remote command list. Use option **2** to change a remote command list. Use option **4** to delete a remote command list and its associated commands. Use option **6** to print a list or ***F21*** to print a full listing of all commands.

After you have created a remote command list, you can use option **12** to add entries to the list. Each entry is one command that can be executed with the **RCMD FTP** subcommand. See the discussion below on the types of entries you can make for IBM i commands.

Add/Change FTP Remote Command List

When you select ***F6*** to add a new list, or option **2** to change a remote command list, the following panel is displayed.



Remote command list name

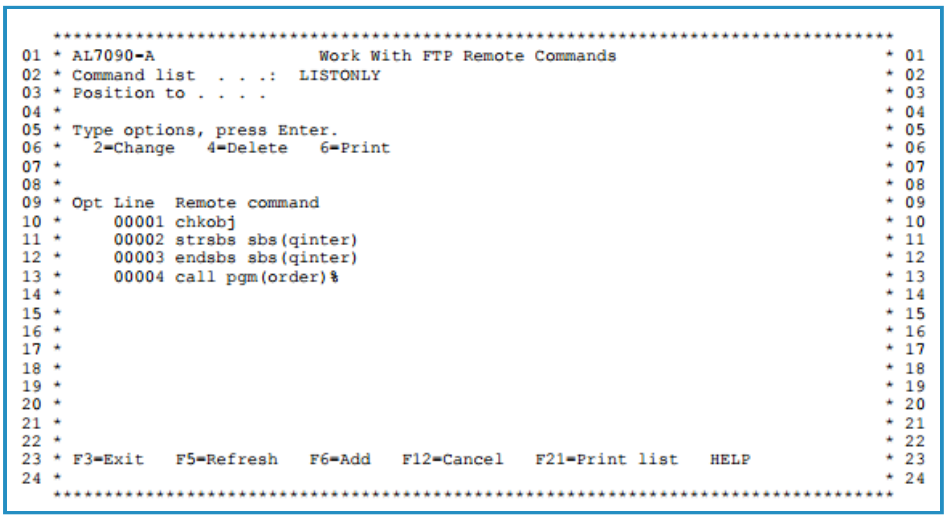
Enter the name of a **`Remote command list`**. The name can have imbedded blanks. This is a required field.

Description

Enter a description for this list.

Work With FTP Remote Commands

This panel is displayed when you use option **`12`** from the **`Work With Remote Commands`** list panel. It will display a list of commands that you are including in the remote command list. If a user is restricted to a remote command list, only those commands that you define here are allowed for that user. For example, if you wanted to give a user access to the **`CALL`** and **`CHKOBJ`** commands, you would need to add two entries to the list - one for each command.



Use ***F6*** to create a new remote command. Use option **`2`** to change a remote command. Use option **`4`** to delete a remote command. Use option **`6`** to print a list or ***F21*** to print a full listing of all commands.

Add/Change FTP Remote Command

When you use ***F6*** to create a new remote command, or option ***2*** to change a system command, the following panel is displayed. From this panel you can enter a command that you want to make available to a user.

Please note the following conventions:

When checking a user-entered command, Alliance checks the entire command against this file. Therefore, if you enter a command as **`chkobj`** and the user enters **`chkobj qgpl/myfile *file`** the command request will be rejected. There must be a match between the command entered and the command in the command list. This comparison is case sensitive. If you enter **`chkobj`** in the command list and the user enters **`CHKOBJ.`** the command will be rejected.

In some cases you may want to check just the first part of a command string rather than the entire command. For example, you may want to give users the ability to use the **`CALL`** command and pass a program variable parameters. If you terminate the command string with the percent character (**`%`**), Alliance will only check the command to that point. For example, if you enter **`call%`** as the command, Alliance will allow the command **`call pgm(invoices)`** to be executed.



Remote command

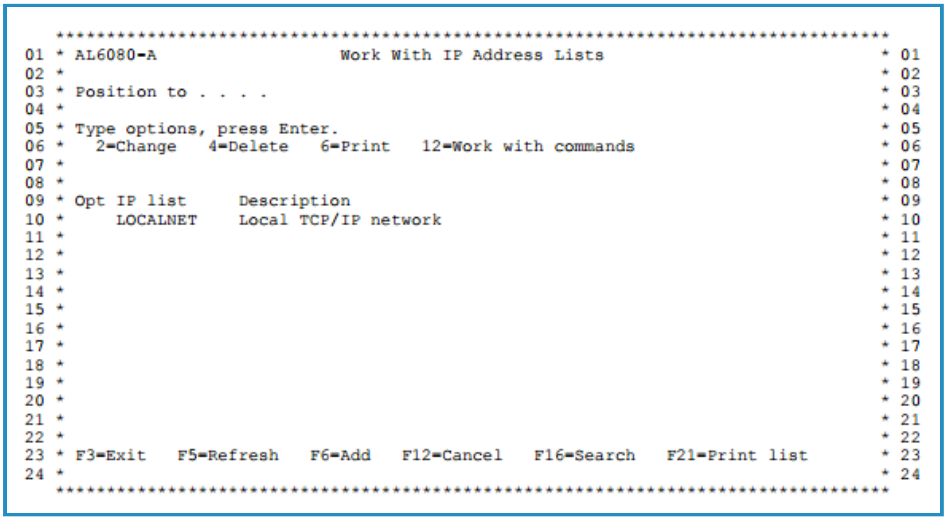
Enter the **`Remote command`** that you want to allow in this list of system commands.

Work With IP Address Lists

IP address checking can be done at two levels: 1) Globally, to check all users who are accessing FTP on your IBM i, and 2) By user, to check the address for a particular user. You define the global IP address list with the Configure FTP Security option from the main menu. You define a user's IP address in the **`Work With FTP Users`** option from the main menu.

When you specify a global IP address list, only FTP users whose computer system has an IP address that is in your list will be allowed to use IBM i FTP. When you specify an IP address list for a user, that particular user can access IBM i FTP only when they are working on their own computer system. This gives you better control over who accesses your system. For example,

if an unwelcome user guesses a user ID and password on your system, you can still reject their logon attempt by enforcing IP address control. If JSMITH works at a computer with an IP address of 147.130.4.17 you can reject this user if they log on from any other system.



Use ***F6*** to create a new IP address list. Use option `2` to change an IP address list. Use option `4` to delete a list and its associated IP addresses. Use option `6` to print a list or *_*F21*_* to print a full listing of all addresses. Use option `12` to work with IP addresses included in a list.

Add/Change IP Address List

When you use ***F6*** to add an address list, or option `2` to change a list, the following panel is displayed. This panel gives you the ability to change the attributes of the list. In this case, you can only change the name and description of the list.



IP address list name

Enter a name for this list. This is a required field and may contain blanks.

Description

Enter a description for this list.

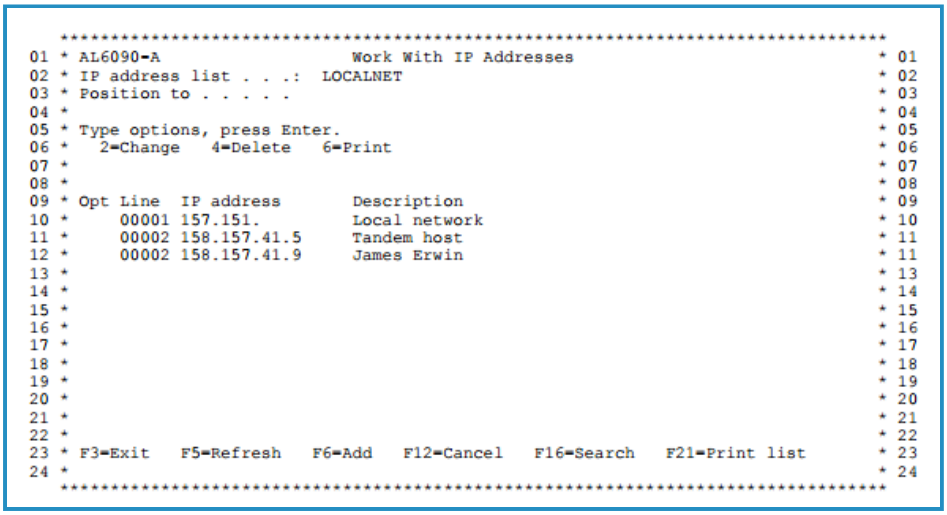
Work With IP Addresses

After you have created a list you can add IP addresses to the list. You should never use an IP address list that has no addresses associated with it. If you use an empty list, the logon attempt will always fail as an address will never be found in the list.

Please note that you can use a portion of an IP address rather than the complete address. This gives you the ability to define all users at a particular network or location. For example, if you have an IP address assigned based on location:

```
151.241.181.1 through 151.241.181.254   New York
151.241.182.1 through 151.241.182.254   Chicago
151.241.183.1 through 151.241.183.254   San Francisco
```

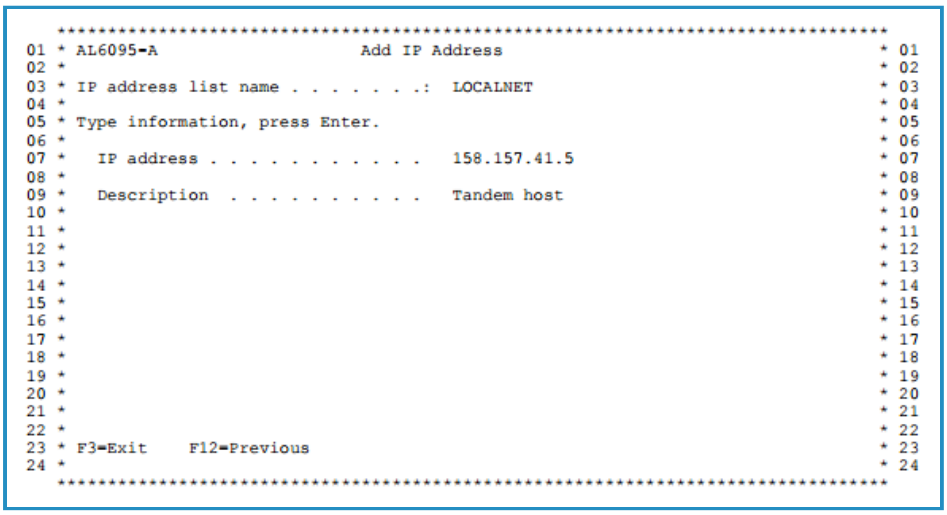
you could enter the IP address “151.241.” to authorize any user in these locations to access FTP, but no one else. Or, if you just wanted to authorize users at New York to access FTP, you could specify IP address “151.241.181.”



Use ***F6*** to add a new address to the list. Use option `2` to change an address. Use option `4` to delete an address from the list. You can print an individual item with option `6`, or the entire list with ***F21***.

Add/Change IP Address

When you use ***F6*** to add a new address, or option `2` to change an address, the following panel is displayed.



IP address

Enter the IP address or partial IP address to add to the list. Use the dotted decimal format for the address. Do not use leading zeros in the network or node portions of the address. For example, enter:

```
128.1.27.4
```

not

```
128.001.027.004
```

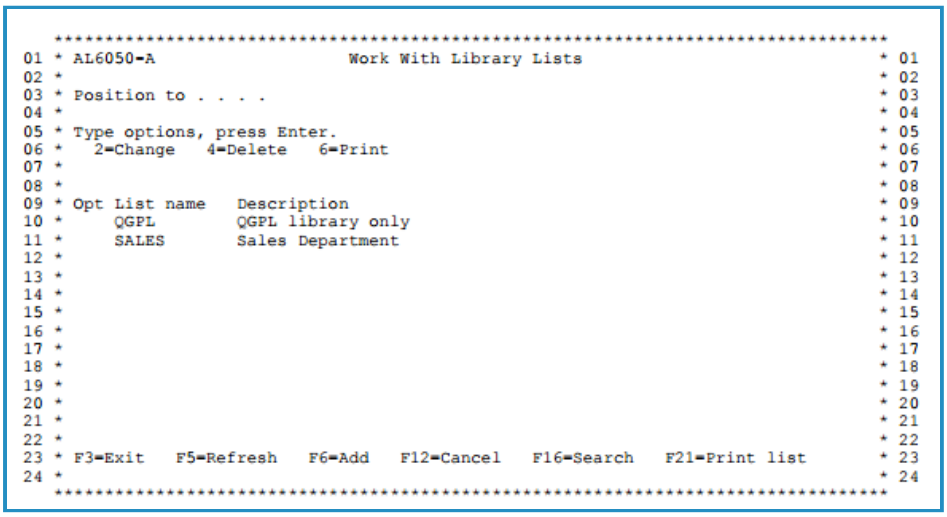
Description

Enter a description for this address. This might be a user name or network name, or any other description you want to give this entry.

Work With Library Security

You can associate a named list of up to 10 libraries that an FTP user can access and Alliance will restrict that user's activities to the list of libraries. Use this option to create the list of libraries.

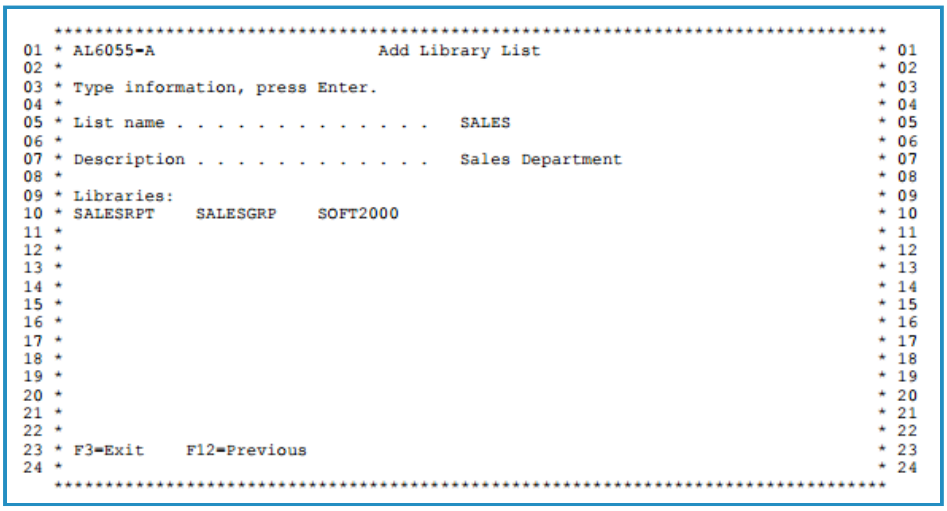
You may wish, for example, to create a library list named **`ANONYMOUS`** and associate this with your anonymous FTP user definition. Anyone accessing FTP with the anonymous ID will be restricted to this library. If a library list is associated with an FTP user ID, the restriction will be applied to all activities that work on libraries or files in libraries.



Use ***F6*** to add a new library list. Use option `2` to change a library list. Use option `4` to delete a library list (the actual libraries will not be affected). You can print an individual library list with option `6`, or the entire set of lists with ***F21***.

Add/Change Library List

If you use ***F6*** to add a new list, or option `2` to change a list, the following panel is displayed. You can add libraries or change the list of libraries.



List name

Enter a name for this list. The name can be up to 10 characters in length and may include imbedded blanks.

Description

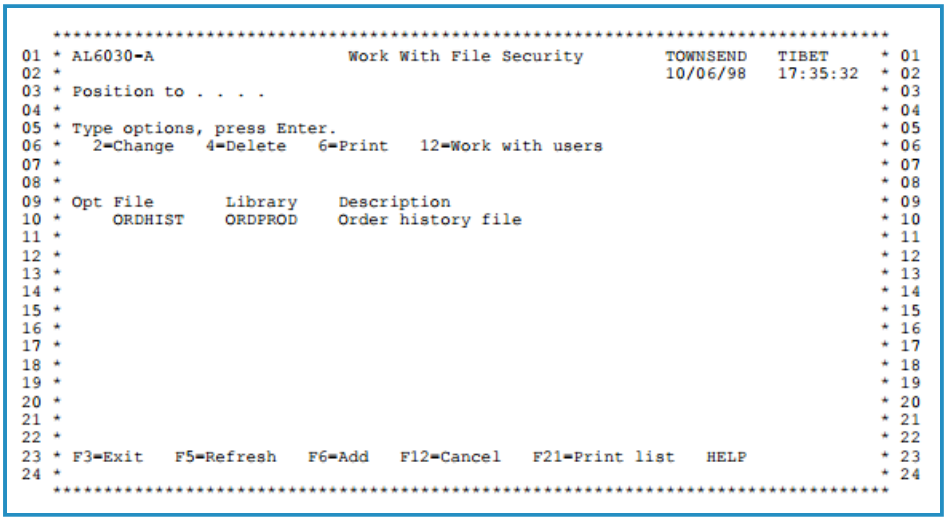
Enter a description for this library list.

Libraries

Enter up to 10 libraries to be included in this list. When a user is validated for access to a library, the list is searched from the first library to the last, excluding blank fields. You should put the most frequently used libraries first in the list for a small performance improvement.

Work With File Security

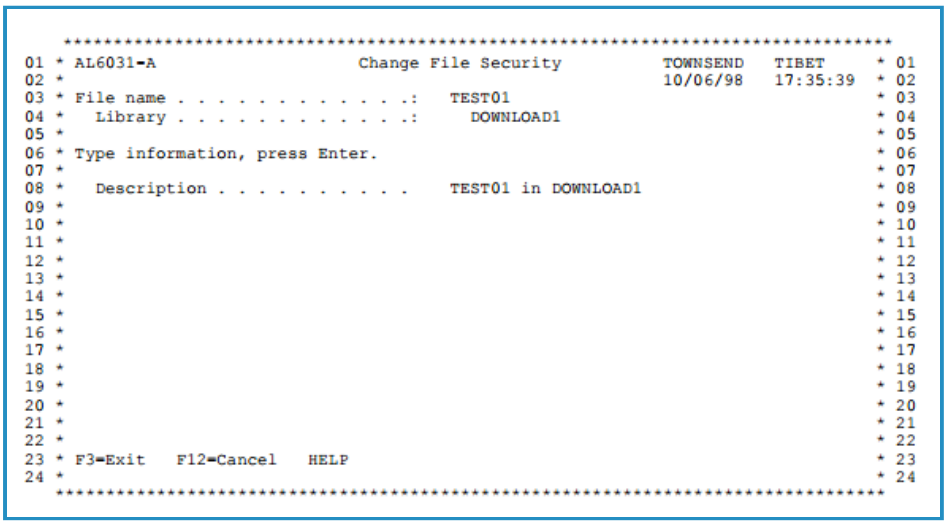
You can authorize one or more users to access a specific file using file security. With file security you can provide a more granular approach to FTP security. Library security, described above, lets you restrict users to a set of libraries. But in some cases you may wish to provide access to a file in a library that should be generally available for user access. File security gives you the ability to define a file to Alliance, and then add one or more users to the authorized list of FTP users. You can still restrict the user to upload, download, rename, or delete functions, or a combination of these functions.



Select option `2` to change the description for this file. Select option `4` to remove this file from Alliance file security. Select option `6` to print a report on security authority for this file. Select option `12` to work with users authorized to access this file. To add a new file use function key ***F6***.

Add/Change File Security

If you use option `2` to change file security, or function key ***F6*** to add a new file, the following panel is displayed.



File name

Enter the name of the file for which you want to establish security. The file does not need to exist at the time you establish security for it.

Library name

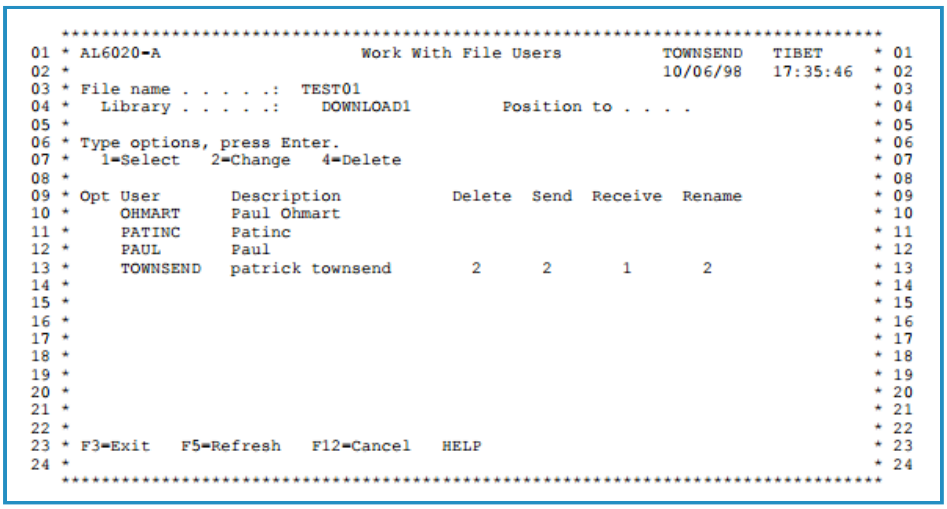
Enter the name of the library which contains the file.

Description

Enter a description for this file.

Work with file users

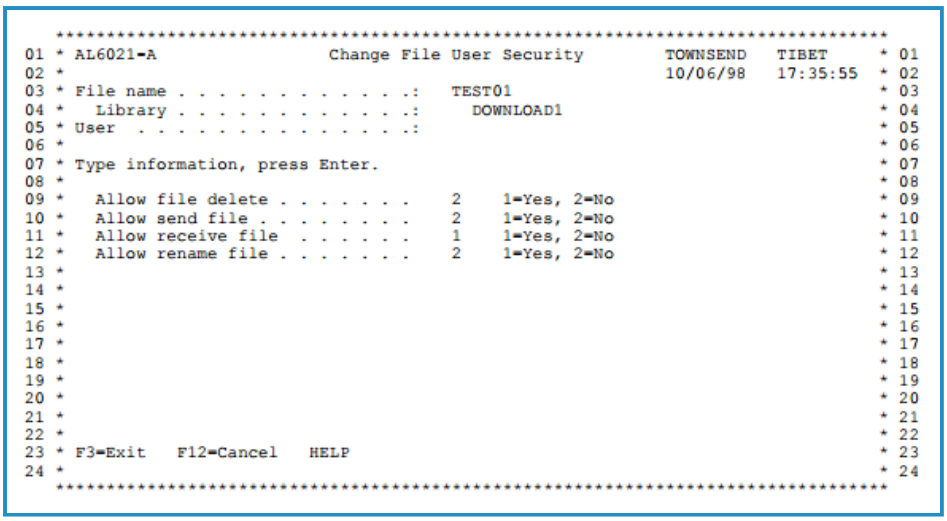
If you use option `12` to work with authorized users for a file the following panel is displayed. You can add one or more users, change the authority of the user, or remove user authority to a file.



Use option `1` to select a user to be added to the security matrix for this file. The user will be added with only the authority to download the file. Use option `2` to change the user's authority to the file. Use option `4` to remove the user's authority to access the file using FTP.

Add/change File User Security

If you use option **2** to change a user's authority for a file the following panel is displayed. You can select the FTP operations this user can perform. Note that you can prevent the user from any of the FTP operations by setting all of the options to **2** for **No**. This is a valid setting and is equivalent to deleting file authority for the user.



Allow file delete

Enter **1** for **Yes** if you want the user to be able to delete this file. Enter **2** for **No** if you do not want the user to be able to delete the file. **No** is the default.

Allow send file

Enter **1** for **Yes** if you want this user to be able to upload this file. Enter **2** for **No** if you do not want the user to be able to upload the file. The default is **No**.

Allow receive file

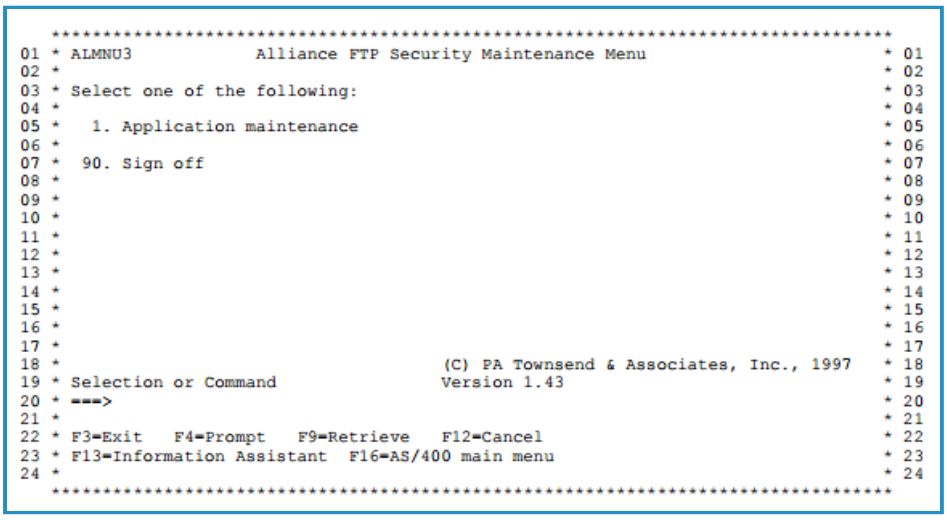
Enter **1** for **Yes** if you want this user to be able to download this file. Enter **2** for **No** if you do not want this user to download the file. The default is **Yes**.

Allow rename file

Enter **1** for **Yes** if you want this user to be able to rename this file. Enter **2** for **No** if you do not want the user to rename the file. The default is **No**.

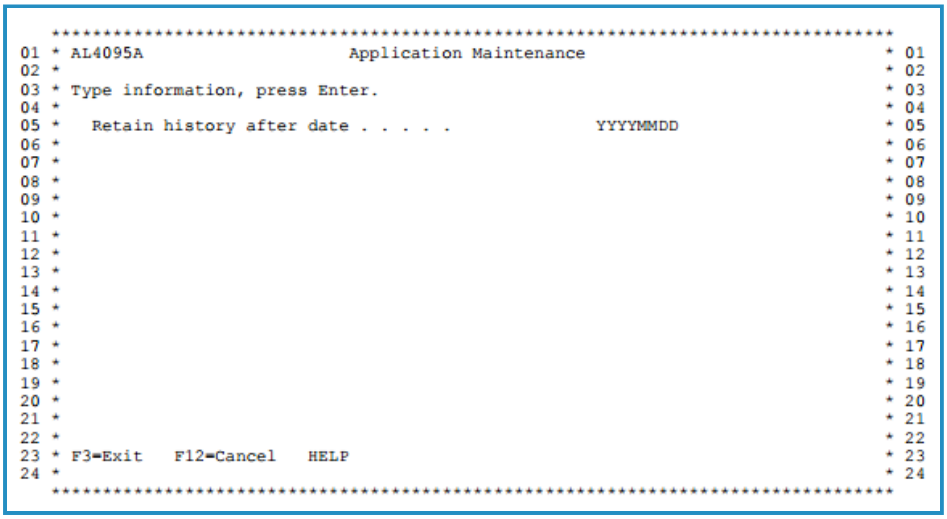
Maintenance

Use this menu to perform routine application maintenance. History and log files will be purged and internal database files will be re-organized. You should perform maintenance weekly or monthly depending on the level of use of the FTP application. Select an option to begin work.



Application Maintenance

This module is used to remove records from Alliance history and logging files and to reorganize database files to recover database space. It is recommended that you run maintenance weekly or monthly depending on the level of use of the security application.



Retain history after date

Enter the date from which you wish to retain history records. The date is entered in **YYYYMMDD** format. Records may be removed from **AFHIST** (history file) and **AFLOGA** (application log file).

Inquiry

Use this menu to access the inquiry functions of Alliance FTP. You can display the application log, work with active FTP jobs, and display FTP access and usage history. Select an option to begin work.

```
ALMNU4                Alliance FTP Security Inquiry Menu

Select one of the following:

   1. Display application log
   2. Work with FTP jobs
   3. Display FTP history

  90. Sign off

                                    (C) PA Townsend & Associates, Inc., 1997
Selection or Command                Version 1.43
===>

F3=Exit    F4=Prompt    F9=Retrieve    F12=Cancel
F13=Information Assistant  F16=AS/400 main menu
```

Display Application Logs

You can use this option to view a list of user actions and the results as recorded by Alliance. From this panel you can also view the details of an entry and print the log. It is important to realize that a single FTP session will generate many log entries. For example, the logon will create one entry, and every request the user makes (change directory, list directory, etc.) will also create a log entry. The application log is intended to provide the system administrator with diagnostic information about the security module, and should only be created when there is a need for such information.

Application log entries will only be created if you enable logging with the **`Configure FTP Security`** option from the main menu. Entries are removed from the log when you use the **`Maintenance`** option on the main menu and specify a date for removing old records. You can also clear the application log with the **`CLRPFM`** command:

```
clrpfm afloga
```

```
AL0360-A                 Work With Application Logs                5/20/97
                                                                   15:18:47
Position to . . . .

Type options, press Enter.
  5=View      6=Print

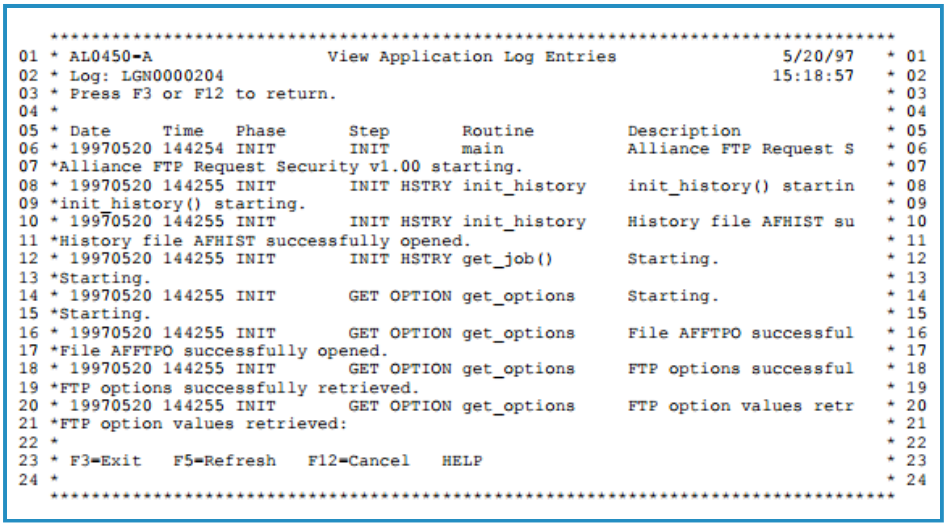
Opt  Log ID       Date       Time     Description
     LGN0000204   19970520   144254   Alliance FTP Request Security v1.00 sta
     LGN0000205   19970520   144302   Alliance FTP Request Security v1.00 sta
     LGN0000206   19970520   144318   Alliance FTP Request Security v1.00 sta
     LGN0000207   19970520   144325   Alliance FTP Request Security v1.00 sta
     LGN0000208   19970520   144358   Alliance FTP Request Security v1.00 sta

F3=Exit    F5=Refresh    F12=Cancel    HELP
```

Enter option `5` to display the details of the log. Enter option `6` to print the log.

Application log detail

When you select option `5` to view the detail on a log the following panel is displayed. You can page forward and back through the log.



```
   ********************************************************************************
01 * AL0450-A                View Application Log Entries              5/20/97   * 01
02 * Log: LGN0000204                                                  15:18:57   * 02
03 * Press F3 or F12 to return.                                                  * 03
04 *                                                                             * 04
05 * Date     Time   Phase      Step       Routine         Description            * 05
06 * 19970520 144254 INIT       INIT       main            Alliance FTP Request S * 06
07 *Alliance FTP Request Security v1.00 starting.                                * 07
08 * 19970520 144255 INIT       INIT HSTRY init_history    init_history() startin * 08
09 *init_history() starting.                                                     * 09
10 * 19970520 144255 INIT       INIT HSTRY init_history    History file AFHIST su * 10
11 *History file AFHIST successfully opened.                                     * 11
12 * 19970520 144255 INIT       INIT HSTRY get_job()       Starting.              * 12
13 *Starting.                                                                    * 13
14 * 19970520 144255 INIT       GET OPTION get_options     Starting.              * 14
15 *Starting.                                                                    * 15
16 * 19970520 144255 INIT       GET OPTION get_options     File AFFTPO successful * 16
17 *File AFFTPO successfully opened.                                             * 17
18 * 19970520 144255 INIT       GET OPTION get_options     FTP options successful * 18
19 *FTP options successfully retrieved.                                          * 19
20 * 19970520 144255 INIT       GET OPTION get_options     FTP option values retr * 20
21 *FTP option values retrieved:                                                 * 21
22 *                                                                            * 22
23 * F3=Exit   F5=Refresh   F12=Cancel   HELP                                   * 23
24 *                                                                            * 24
   ********************************************************************************
```

Date

For each line, this is the date the line was created in **`YYYYMMDD`** format.

Time

For each line this is the time the line was created in **`HHMMSS`** format.

Phase

The phase describes the high level activity of the security module at the time the log was written.

Step

The step describes the activity within a phase.

Routine

The routine records the program routine in which the log entry was created.

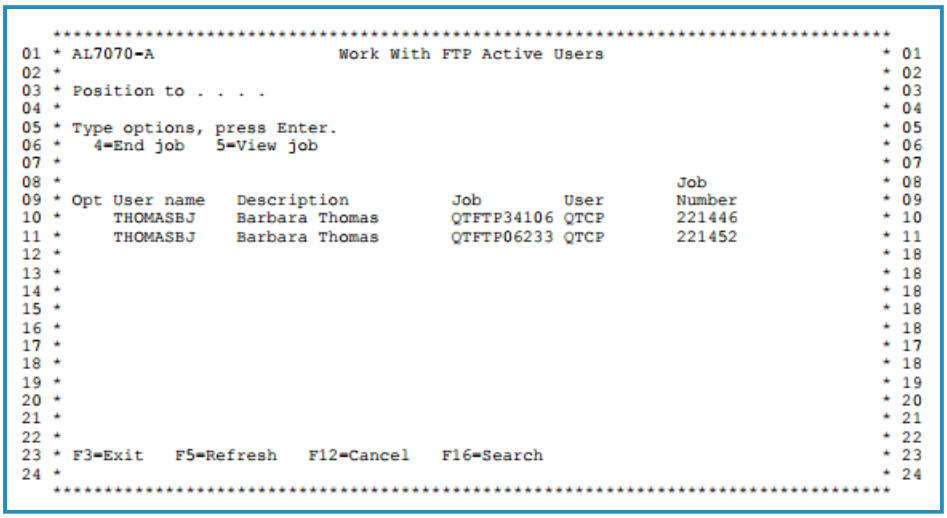
Description

The description provides diagnostic information about the activity and/or errors encountered by the Alliance program. The reasons for rejecting a request will appear in the descriptive text.

Work With FTP Jobs

When Alliance accepts a logon request for an FTP user it writes an entry to the active jobs file. You can display a list of FTP users on your system with this option. If the user exists in the Alliance FTP Users file the name of the user will be shown in the description field.

You can display the FTP server job with option 5 of this panel. You can also end an FTP server job with option **`4`** . However, you should be careful when using option **`4`** as you will end the FTP server immediately.

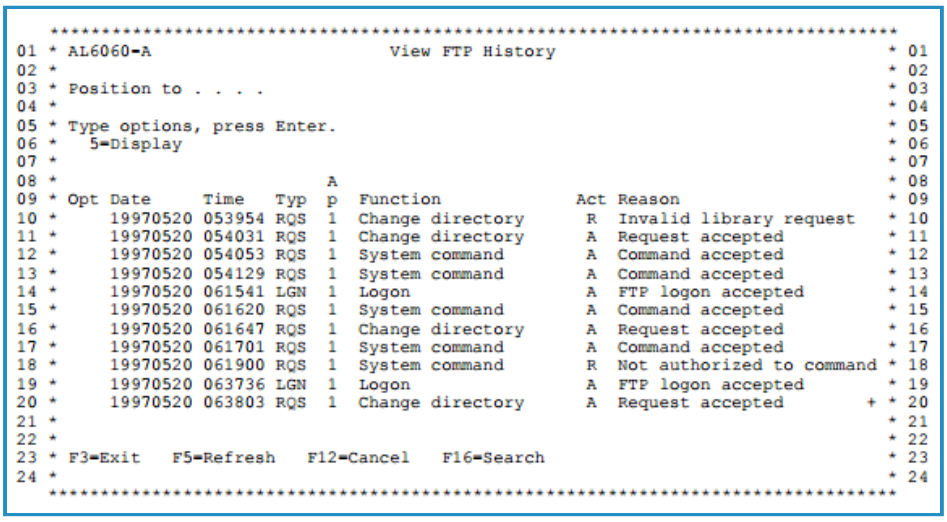


Use ***Page Up*** or ***Page Down*** to move through the list of users. Use option **`5`** to view the job for a specified user. Use option **`4`** to end the FTP job for the user.

Display FTP History

This module provides a summary view of the status of all logon and command requests made with FTP. It shows the date, time, type (**`Logon`** or **`Request`**), application, function, action taken (**`Accept`** or **`Reject`**) and reason for the action. The system administrator can use this panel to view the history of activity and to assist in problem determination.

The history file is **`AFHIST`** and is an externally described file. The file can be queried using any file utility or query product such as IBM Query, etc. Records are removed from this file only when the Maintenance option is run from the main menu. Records are removed based on the date entered from that panel.

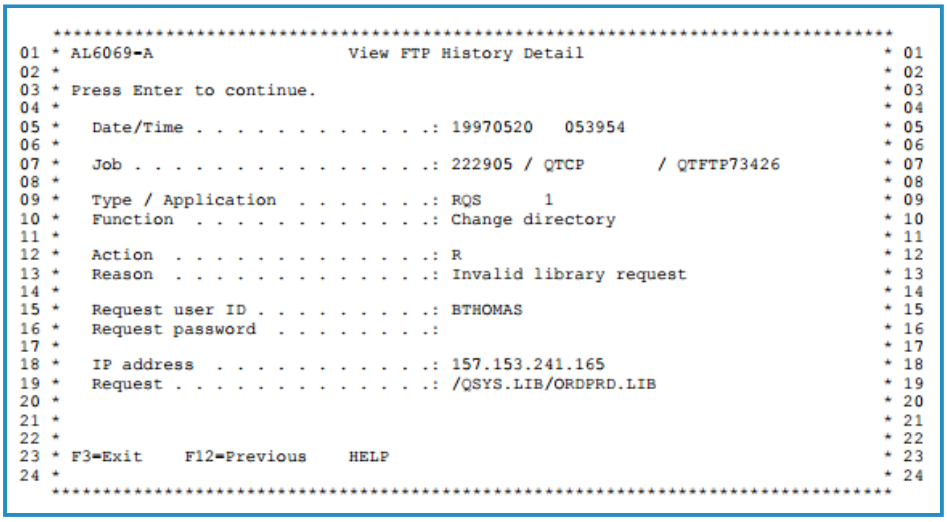


```
01 * AL6060-A                        View FTP History                          * 01
02 *                                                                           * 02
03 * Position to . . . .                                                      * 03
04 *                                                                           * 04
05 * Type options, press Enter.                                               * 05
06 *   5=Display                                                               * 06
07 *                                                                           * 07
08 *                            A                                              * 08
09 * Opt Date     Time   Typ  p  Function             Act Reason                * 09
10 *     19970520 053954 RQS  1  Change directory      R  Invalid library request * 10
11 *     19970520 054031 RQS  1  Change directory      A  Request accepted      * 11
12 *     19970520 054053 RQS  1  System command        A  Command accepted      * 12
13 *     19970520 054129 RQS  1  System command        A  Command accepted      * 13
14 *     19970520 061541 LGN  1  Logon                 A  FTP logon accepted    * 14
15 *     19970520 061620 RQS  1  System command        A  Command accepted      * 15
16 *     19970520 061647 RQS  1  Change directory      A  Request accepted      * 16
17 *     19970520 061701 RQS  1  System command        A  Command accepted      * 17
18 *     19970520 061900 RQS  1  System command        R  Not authorized to command * 18
19 *     19970520 063736 LGN  1  Logon                 A  FTP logon accepted    * 19
20 *     19970520 063803 RQS  1  Change directory      A  Request accepted    + * 20
21 *                                                                           * 21
22 *                                                                           * 22
23 * F3=Exit   F5=Refresh   F12=Cancel   F16=Search                            * 23
24 *                                                                           * 24
```

Use option `6` to view the details for any history line.

Display history detail

Use this panel to view the details for a history log entry.



```
01 * AL6069-A                  View FTP History Detail                       * 01
02 *                                                                         * 02
03 * Press Enter to continue.                                               * 03
04 *                                                                         * 04
05 *   Date/Time . . . . . . . . . . . .: 19970520   053954                  * 05
06 *                                                                         * 06
07 *   Job . . . . . . . . . . . . . . .: 222905 / QTCP       / QTFTP73426   * 07
08 *                                                                         * 08
09 *   Type / Application  . . . . . . .: RQS      1                       * 09
10 *   Function  . . . . . . . . . . . .: Change directory                 * 10
11 *                                                                         * 11
12 *   Action  . . . . . . . . . . . . .: R                                * 12
13 *   Reason  . . . . . . . . . . . . .: Invalid library request          * 13
14 *                                                                         * 14
15 *   Request user ID . . . . . . . . .: BTHOMAS                          * 15
16 *   Request password  . . . . . . . .:                                  * 16
17 *                                                                         * 17
18 *   IP address  . . . . . . . . . . .: 157.153.241.165                  * 18
19 *   Request . . . . . . . . . . . . .: /QSYS.LIB/ORDPRD.LIB             * 19
20 *                                                                         * 20
21 *                                                                         * 21
22 *                                                                         * 22
23 * F3=Exit    F12=Previous    HELP                                         * 23
24 *                                                                         * 24
```

Date / time

The date and time field records the date and time of the log record. Date is in **`YYYYMMDD`** format and time is in **`HHMMSS`** format.

Job

The job number, user name, and job name are recorded for the history record.

Type / application

There are two types of log entries: LGN is a logon request and RQS is an FTP command request. The application ID is `1` for server and `0` for client. Alliance only provides security support for the server application, so the application will always be `1` for server.

Function

The function field describes the user activity that was requested. This could be a request to change the default library, upload or download a file, or make a file list.

Action

The action field describes whether this request was accepted (**`A`**) or rejected (**`R`**). The following field provides a brief description of the reason for the action taken.

Reason

A brief description of the action taken. If the action was to reject the request, this field will contain information on why the request was refused.

Request user ID / password

The request user ID is the user ID of the client. This information is always provided. The password is the password entered during a connection to the IBM i FTP server. The password is only available for the Logon transaction - this field is blank for the FTP request record. If the user ID is ANONYMOUS this field will contain the e-mail address of the client, by convention.

IP address

This field contains the IP address, in dotted format, of the requesting user.

Request

The request field contains information about the transaction. If the request was to change the default directory, this will have the directory for the change. If the request is to upload, download, or rename a file, this field will contain the file name. File names are always in long format. For example, file **`ORDERS`** in library **`ORDPRD`** would be listed as:

```
/QSYS.LIB/ORDPRD.LIB/ORDERS.FILE/ORDERS.MBR
```

Support

Use this menu to access all Alliance Support functions. You will normally only use these functions when working with your software vendor to resolve questions or problems on the use of Alliance FTP.

```
                                                                                
01 * ALMNU5              Alliance FTP Security Support Menu                    * 01
02 *                                                                           * 02
03 * Select one of the following:                                             * 03
04 *                                                                           * 04
05 *   1. Start remote support                                                 * 05
06 *   2. End remote support                                                   * 06
07 *                                                                           * 07
08 *  90. Sign off                                                             * 08
09 *                                                                           * 09
10 *                                                                           * 10
11 *                                                                           * 11
12 *                                                                           * 12
13 *                                                                           * 13
14 *                                                                           * 14
15 *                                                                           * 15
16 *                                                                           * 16
17 *                                                                           * 17
18 *                                  (C) PA Townsend & Associates, Inc., 1997 * 18
19 * Selection or Command              Version 1.43                            * 19
20 * ===>                                                                     * 20
21 *                                                                           * 21
22 * F3=Exit    F4=Prompt    F9=Retrieve    F12=Cancel                         * 22
23 * F13=Information Assistant  F16=AS/400 main menu                          * 23
24 *                                                                           * 24
                                                                                
```

Start Remote Support

This option is provided to allow your software vendor to perform problem analysis or software updates. When you use this option a line description, controller description, and device description will be created for a dial-in connection to your IBM i. A user profile named PATINC will also be created. The connection will use your normal ECS modem. Your software vendor will request the phone number of your ECS modem to make the connection.

When software support is complete you should run the **`End Remote Support`** to disable the user profile PATINC and vary the line, controller, and device off.

```
                                                                                
01 * AL4080A                    Start Remote Support                        * 01
02 *                                                                        * 02
03 * Type information, press Enter.                                        * 03
04 *                                                                        * 04
05 *   Resource name  . . . . . . . . . . .   LIN011                       * 05
06 *                                                                        * 06
07 *   Line speed . . . . . . . . . . . . .   9600         2400, 4800, 9600, 14400  * 07
08 *                                                                        * 08
09 *                                                                        * 09
10 *                                                                        * 10
11 *                                                                        * 11
12 *                                                                        * 12
13 *                                                                        * 13
14 *                                                                        * 14
15 *                                                                        * 15
16 *                                                                        * 16
17 *                                                                        * 17
18 *                                                                        * 18
19 *                                                                        * 19
20 *                                                                        * 20
21 *                                                                        * 21
22 *                                                                        * 22
23 * F3=Exit    F12=Cancel                                                   * 23
24 *                                                                        * 24
                                                                                
```

Resource name

Enter the resource name of your ECS line. This is usually **`LIN011`** or **`CMN01`** . You can determine the resource name by displaying the line description **`QESLINE`** :

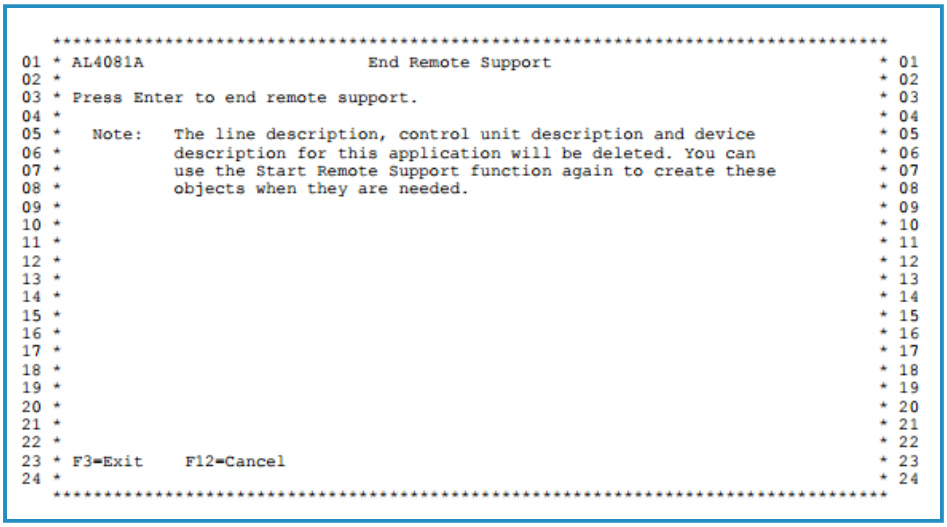
```
dsplind qesline [Enter]
```

Line speed

Enter the speed of your ECS modem. This is usually 9600 baud. For an IBM 5853 modem, the speed is 2400. For an IBM 7855 the speed is 9600. For an IBM 7856 modem the speed is 14400.

End Remote Support

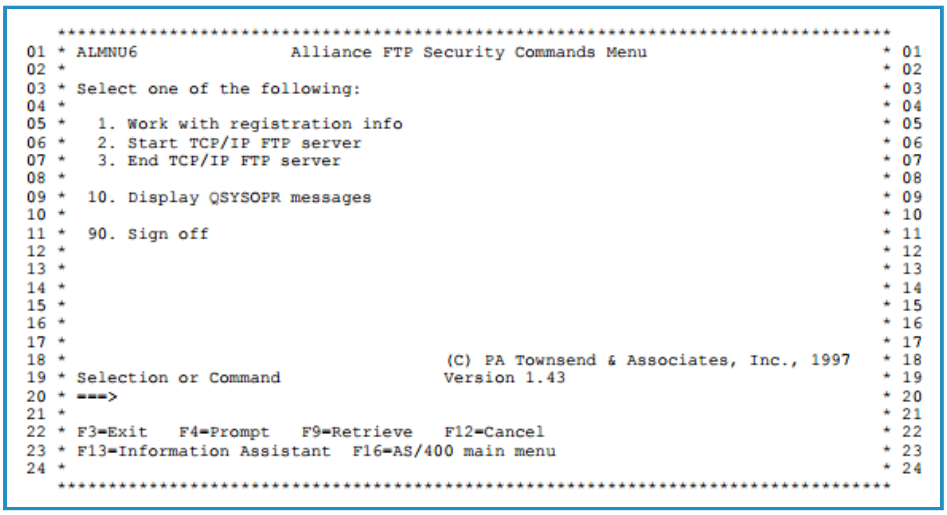
Use this option when your software vendor has finished with remote support. This option will disable the support user profile named PATINC and vary off the line, controller, and device description.



There are no input fields on this panel. Press ***Enter*** to end remote support.

FTP Security Commands

The options on this menu provide access to common IBM i FTP security or configuration commands. They are provided here for convenience. Please refer to the IBM i documentation for a description of these commands.



Getting Support

Free support

Free support is available for the first 90 days after you install Alliance. You can receive telephone support, free fixes to problems, and support in the use of the product.

Send questions via e-mail to: support@townsendsecurity.com

Please be sure to include your name, return e-mail address, company name and address, and phone number. We attempt to answer all questions within 24 hours of receipt.

Send questions via fax to: (360) 357-9047

Please be sure to include your name, return fax number, company name and address, and phone number. We try to answer all questions within 24 hours of receipt.

Paid support

An annual maintenance contract is available that provides you with telephone support, fixes to problems as they are available, new releases and enhancements on a regular basis, and other services. Contact your software vendor for information about the annual maintenance agreement.

Problem determination

Overview

Most problems you will encounter with Alliance will probably have to do with configuring the many options that are available for controlling FTP access. However, you may encounter actual program problems that require support. Please consider using the following procedures to analyze a problem before reporting the problem to your software vendor.

Using the history log

Alliance records the result of every logon and request that is made via FTP for registered users. Use the **Display History Log** option on the main menu to view the results of a logon or request. The history log will provide a short description of any request that was rejected. You may be able to determine why a user request is being rejected directly from the history log.

Using the application logs

For a more detailed description of the actions Alliance is taking during logon or request processing, you can activate the application log. Use the **Configure FTP Security** option to activate logging. You do not need to terminate and restart FTP to make this option active. After activating application logging, replicate the problem and view the log with the **Display Application Log** option on the main menu. The log will have a detailed list of functions performed during logon or request validation. You may be able to determine the cause of the problem by reviewing the log.

FTP job logs

If you are unable to determine the cause of a problem, or if FTP is rejecting your logon or requests but the application log indicates that the logon attempt or request was accepted, you can view the job logs for FTP to try to determine the cause of the problem. To view the job logs you must end FTP server functions. Use the **ENDTCPSVR** command to do this:

```
endtcpsvr *ftp
```

After ending FTP services you can locate the job logs using the **WRKUSRJOB** command:

```
wrkusrjob qtcp
```

FTP server jobs all have the job name in the format **QTFTPnnnnn** where nnnnn is a number assigned to the job. Look for any FTP job log that has more than one page. This usually indicates a potential problem. View the job log to determine if there are any error or diagnostic messages that can be helpful in solving the problem.

QSYSOPR messages

FTP applications rarely place messages in the **QSYSOPR** message queue, but you should check there if you are having difficulties. Use the **DSPMSG** command to view this message queue:

```
dspmsg qsysopr
```

Print any error messages from FTP that may be there.

Further action

If you are unable to determine the cause of a problem, please contact your software vendor for support. Be prepared to provide a copy of the history log, application log, and any **QSYSOPR** messages to your software vendor.